MAHARASHTRA STATE BOARD OF TECHNICAL EDUCATION, MUMBAI

A Project Report on

Traditional Attire Rental Shop Website

Submitted as a partial fulfillment for the award of

Diploma in Computer Engineering

Submitted by

Kunal Raghunath Chaudhari

Sakshi Vinod Chavan

Harshali Vijay Fulpagare

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DEPARTMENT OF COMPUTER ENGINEERING

S.S.V.P.S's Bapusaheb Shivajirao Deore Polytechnic, Dhule

(Institute Code : 0059)

2025 – 2026

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DEPARTMET OF COMPUTER ENGINEERING

CERTIFICATE

This is to certify that,

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has satisfactorily completed project entitled **"Traditional Attire Rental Shop Website"** in partial fulfillment for the **Diploma Course in Computer Engineering** of the **Maharashtra State Board of Technical Education** at our polytechnic during the academic year **2025-2026**. The project is completed by a group consisting of four persons under the guidance of the Faculty Guide.

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Place: Dhule

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Faculty Guide

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Name :
External Examiner

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Name : Mr. N. D. Patel
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Signature:
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Principal.

Namrata
NAUVARI SPECIAL

TO WHOMSOEVER IT MAY CONCERN

This is to certify that the following students of **Final Year Diploma in Computer Engineering** from **S.S.V.P.S's B. S. Deore Polytechnic, Dhule** have successfully completed their **Capstone Project** titled: **"Traditional Attire Rental Shop Website"**

Name of Students:

1. Kunal Raghunath Chaudhari.
2. Sakshi Vinod Chavan.
3. Harshali Vijay Fulpagare.
4. Daksh Pravin Jadhav.

This project was sponsored by our organization and developed as per our requirements.

We appreciate their sincere efforts, dedication, and hard work during the project.

We wish them all the best for their bright future.

For,
Namrata Nauvari Special Shop,
Gurukul Colony, Near Bus Stand,
Taloda, Dist. Nanadurbar-425413

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We would like to extend our heartiest thanks to the management of our college for providing us with the necessary resources, facilities, and support required to complete this project efficiently.

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Kunal Raghunath Chaudhari

Sakshi Vinod Chavan

Harshali Vijay Fulpagare

Daksh Pravin Jadhv

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Abstract

The **"Traditional Attire Rental Shop Website"** is designed to provide an efficient and user-friendly digital platform for renting traditional clothing. In today's fast-paced lifestyle, purchasing expensive traditional outfits for one-time use is not always practical. This system offers a convenient solution where users can easily browse, select, and rent attire based on their preferences, size, and occasion.

The developed system simplifies the entire rental process through features such as user registration, advanced search and filtering, real-time availability tracking, booking management, and secure payment processing. It ensures transparency in pricing by clearly displaying rent, deposit, and additional charges. By maintaining digital records, the system reduces manual errors and improves operational efficiency for both customers and shop owners.

The system is developed using modern web technologies to ensure performance, scalability, and reliability. The frontend is built using **React.js and Vite.js** along with HTML, CSS, and Bootstrap to create an interactive and responsive user interface. The backend is developed using **Node.js** and **Express.js**, which handle server-side logic and API communication. **MongoDB** is used as the database to store user data, product details, bookings, and payment records efficiently. Additionally, **Razorpay payment gateway** is integrated to enable secure and seamless online payment transactions.

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Chapter 1

Introduction

This chapter introduces the concept of the rental management system and provides an overview of the sponsored capstone project. It explains the basic idea of rental business operations, including how products are managed, booked, and returned. The chapter also describes the key activities involved in the system and the overall rental process flow.

The purpose of this chapter is to build a clear understanding of how rental businesses function and how a digital system can improve their efficiency. It highlights the importance of proper management of inventory, bookings, payments, and customer records.

Namrata Nauvari Special Shop:-

The Traditional Attire Rental Shop, named "Namrata Nauvari Special Shop", is located in Taloda, Nandurbar. The shop was established in the year 2019 by Mrs. Namrata Shivade. Before starting her business, she completed professional training from an institute where she learned the skill of stitching Nauvari sarees and gained knowledge in traditional attire design. Prior to 2019, she worked for around 2 to 3 years in tailoring work, mainly focusing on fall and edging of sarees and blouse stitching, which helped her gain practical experience and confidence.

After gaining sufficient skills and experience, she started her own shop specializing in Nauvari sarees and traditional attire. The shop provides rental services for Nauvari sarees, lehengas, traditional blouses, and matching jewellry. The main aim of the shop is to offer affordable and attractive traditional outfits to customers for various occasions without the need to purchase expensive attire. Due to her dedication and consistent efforts, the shop has grown steadily and gained popularity among customers in the local area. Today, it serves customers for weddings, festivals, cultural programs, and other traditional events, providing quality service and customer satisfaction

Rental Work

The rental business is the process of providing products to customers for temporary use in exchange for rent. It acts as a service-oriented business model where ownership of the product remains with the rental provider, while customers use items for a specific period. Rental services can be managed through physical stores, online platforms, or a combination of both.

In daily life, rental services are very useful because they provide affordable and flexible access to products without full purchase cost. Customers can rent items as per their need, duration, and budget. Rental businesses also support the economy by creating employment, improving resource utilization, and reducing unnecessary consumer spending.

The rental business involves several key activities and processes to ensure smooth operations and customer satisfaction. These include:

Product and Category Management: Rental providers must maintain proper list of products with category-wise details so customers can easily find suitable items. Correct classification helps in quick booking and better stock visibility.

Booking Management: Every booking must record customer details, product details, rental duration, and booking status. Proper booking management avoids conflicts and double allocation of items.

Availability and Inventory Tracking: Real-time tracking of product availability is essential. It helps staff know which items are currently rented, which are available, and which are under inspection or return process.

Inspection and Return Handling: Inspection is performed before delivery and at the time of return to ensure product quality and accountability. Any damages, delays, or issues in product condition are carefully recorded, and appropriate corrective or penalty actions .

Payment Management: Rental businesses require accurate payment handling, including advance payment, pending amount, and final settlement. Proper payment records improve transparency and financial control.

Customer Service: Good customer support improves trust and repeat business. This includes assisting customers during booking, handling changes in rental period, and resolving complaints quickly.

To ensure effective service and business growth, the rental process generally follows these steps:

Step 1: Product Listing and Classification

Products are listed with category and details so that staff and customers can identify items quickly.

Step 2: Customer Registration and Booking

Customer information is recorded, and booking is created based on product availability and rental duration.

Step 3: Verification and Delivery

Before issuing the product, booking details are verified and item condition is checked.

Step 4: Payment Collection

Rent and related charges are collected through suitable payment modes and recorded in the system.

Step 5: Return and Inspection

After rental period completion, the product is returned and inspected for condition, late return, or damages.

Step 6: Final Settlement and Reporting

Any pending payment or charges are settled, and records are updated for future analysis and decision-making.

Table 1.1 Rental Process Summary Table

Step No.	Process Name	Description
1	Product Listing	Products are categorized and listed properly
2	Customer Booking	Customer details and booking information are recorded

Step No.	Process Name	Description
3	Verification & Delivery	Product condition is checked before delivery
4	Payment Collection	Rent amount and charges are collected
5	Return & Inspection	Product is returned and inspected for damages
6	Final Settlement	Final payment is completed and records are updated

This chapter introduced the rental management system and explained the concept of the rental business. It covered the key activities involved in rental operations and described the overall rental process flow. The understanding developed in this chapter provides a foundation for designing an efficient rental management system.

Chapter 2

Literature Survey

This chapter presents the study of existing rental systems used in traditional attire rental businesses by "Namrata Nauvari Special Shop", is located in Taloda, Nandurbar. It focuses on understanding how current systems operate, including manual processes and online rental platforms. The chapter highlights the limitations and problems faced in these systems, which affect efficiency and customer satisfaction.

The purpose of this study is to analyze real-world issues such as booking conflicts, payment handling problems, and lack of proper record management. This analysis helps in identifying the need for an improved rental management system.

2.1 Requirement Gathering

Questions Asked in Owner:

Q1. Do you face issues like double booking due to manual record updates?
Answer: Yes, sometimes double booking happens because records are maintained manually and not updated properly.

Q2. Do you experience delays in the booking confirmation?

Answer: Yes, booking confirmation takes time because availability needs to be checked manually.

Q3. Is there any confusion in pricing, rent, deposit, or extra charges?
Answer: Yes, sometimes customers get confused about total charges as everything is not clearly calculated.

Q4. Do you face problems in managing payments or tracking transactions?
Answer: Yes, it is difficult to track advance payments, pending amounts, and final payments manually.

Q5. Is it difficult to track booking history and previous records?

Answer: Yes, searching old records in registers takes time.

Q6. Do you have a proper complaint handling or customer support system?
Answer: No, there is no proper complaint system, issues handled manually.

Q7. Do you find it difficult to check product availability in real-time?

Answer: Yes, availability is checked manually, so it is not always accurate.

Q8. Is managing customer details manually time-consuming?

Answer: Yes, writing and maintaining customer details manually takes a lot of time.

Q9. Do you face issues while handling returns and inspection records?

Answer: Yes, inspection details are not recorded properly, which creates confusion after return.

The shop owner faces several challenges such as double booking, delay in booking confirmation, payment tracking issues, and difficulty in managing records. The absence of a proper digital system leads to errors and inefficiency. Therefore, there is a need for a system that can automate these processes and improve accuracy.

2.2 Study of Existing system

This section presents the study of existing rental systems used in traditional attire rental businesses. These systems include manual processes as well as partially digital and online platforms. The study focuses on understanding their working methods, features, and limitations.

2.2.1 Manual Process System

Currently, most traditional attire rental shops operate using a manual system. In this system, booking details, customer information, payment records, and return dates are maintained in paper registers or simple spreadsheets. Communication with customers is carried out through phone calls or messages. This system is highly dependent on manual work, which increases the chances of errors and reduces overall efficiency.

Key Issues in Manual System:

- High chances of human error in record keeping
- Difficulty in searching old booking records
- No real-time availability tracking

- Risk of data loss due to damage or misplacement
- Time-consuming process
- No proper tracking of payments and booking history

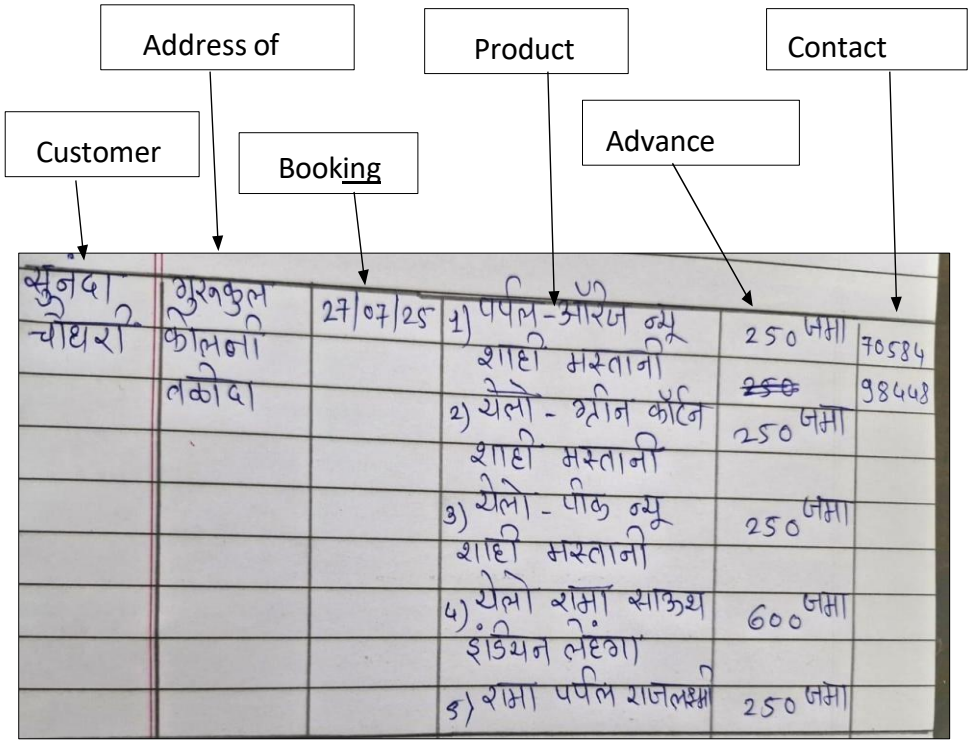


Figure 2.1 Manual Booking Record

As shown in Fig. 2.1, booking records are maintained manually using registers, which makes data management difficult and time-consuming.

The existing system stores customer and booking details manually in registers, which makes data management difficult and error-prone. There is no proper structure to organize or verify information, leading to confusion, inefficiency, and chances of data loss. It also becomes hard to track bookings and payments accurately, especially when multiple requests are handled at the same time.

Customer Details:

- Stored manually (name, contact, address)
- Difficult to search, update, and verify
- Possibility of incorrect or incomplete data
- No proper data security or validation

Booking Details:

- Recorded manually (items, date, duration)

• No proper booking status tracking

• High chances of errors and missing entries

• Risk of double booking due to no real-time updates

• Difficult to manage multiple bookings simultaneously

2.2.2 Product Categories Structure

The product categories in the existing rental system are organized into different groups to make searching and selection easier for customers. The main categories include:

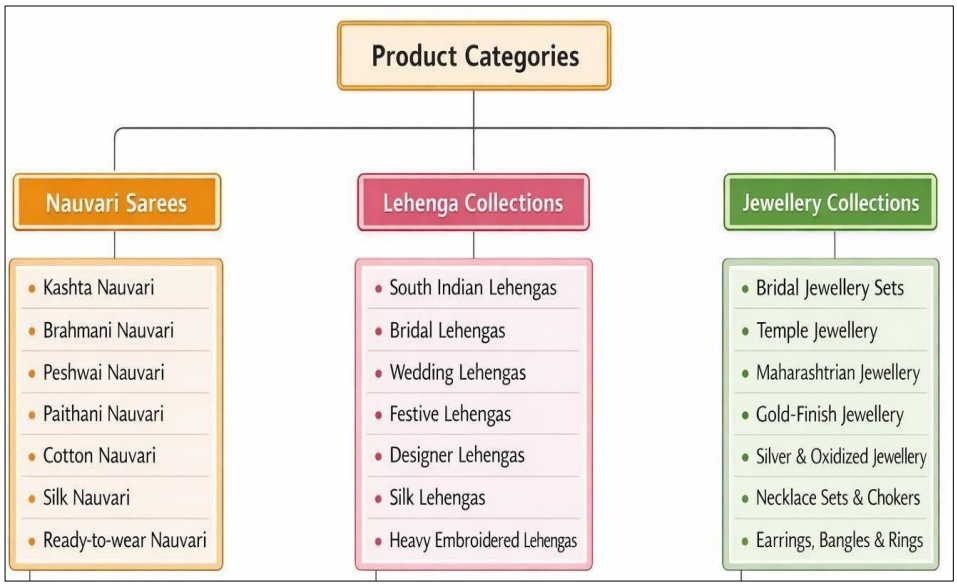


Figure 2.2 Product Categories Structure

Products in the existing rental system are organized into categories such as Nauvari sarees, lehengas, and jewellery. Each category is further divided into subcategories based on type and occasion. This structure helps customers easily search and select suitable products. It also improves organization and efficiency of the rental system.

- Products are divided into main categories for better organization
- Subcategories are created based on type and occasion
- Makes product searching easy and fast
- Reduces confusion while browsing
- Improves navigation and user experience
- Helps in better inventory management
- Supports handling of large product collections

- Allows easy comparison of similar items
- Saves time for customers and administrators
- Increases overall system efficiency

2.3 Limitations of Existing Systems

Based on the study of existing platforms, the following table summarizes the key limitations and problems identified across major functional areas.

Table 2.1: Limitations of Existing Rental Systems

Sr. No.	Area	Limitations
1	Availability	Double booking due to manual updates
2	Booking Approval	Delay in confirmation
3	Pricing Transparency	Lack of clarity in rent + deposit + extra charges breakdown
4	Payment Handling	Refund issues, payment failures, and unclear transaction records
5	Record Keeping	Poor tracking of booking history and absence of payment proofs
6	User Trust	Fake or poor quality products
7	Customer Support	No complaint system

2.4 Problem Identification

After studying the existing systems, the following problems were clearly identified for both users and owner/admin stakeholders. These issues highlight the limitations of manual processes and lack of system automation. They also indicate the need for a more efficient, accurate, and user-friendly web-based rental management system.

Table 2.2: Problem Identification for Users and Owner/Admin

Stakeholder	Problem Identified
User Problems	Difficulty in finding suitable rentals quickly due to weak search filtering.
	Confusion in pricing due to hidden charges or unclear deposit/refund rules.
	Booking failures caused by availability mismatch (double booking).
Owner/Admin Problems	Manual handling of booking requests leads to delays and errors.
	Difficulty in managing cancellations, refunds, and disputes.
	Lack of a complete digital record system increases risk of disputes.

2.5 Needs of system

Based on the problems identified above, the following table outlines the functional requirements and needs of the proposed Rental Application system:

Table 2.3: Need of System

Sr. No.	Role	Need of System
1	User	Easy search & filter (location, price, category, date)
2	User	Real-time availability tracking to prevent double booking

Sr. No.	Role	Need of System
3	User	Transparent pricing with automated rent + deposit + charge calculations
4	Owner and Admin	Easy-to-use booking and listing management dashboard
5	Owner and Admin	Digital record system for all bookings payment history
6	Owner and Admin	Strong verification mechanism and dispute handling support

2.6 Problem Definition

Managing a rental business through a web application is challenging because the system must track listings, availability, bookings, and payments accurately. Problems like double booking, delayed approvals, and unclear records can reduce efficiency and cause confusion. Therefore, a Rental Web Application is needed to provide real-time availability, proper booking tracking, and digital records, making the system simple, accurate, and easy to manage.

This chapter analyzed the existing rental systems and identified their major limitations, including manual errors, lack of real-time tracking, and inefficient record management. It also highlighted the key problems faced by both users and administrators. Based on this study, the need for a structured and digital rental management system was clearly established.

Chapter 3

Scope of the Project

This chapter defines the scope of the Traditional Attire Rental Shop Management System and outlines the functionalities included in the project. It explains how the system replaces manual rental operations with a structured digital approach.

The objective of this chapter is to clearly define what the system will do and what is excluded from the project.

The system is designed to replace manual handling of rental operations such as product tracking, booking, return, inspection, and payment recording.

The scope focuses on:

- Improving day-to-day rental operations
- Providing real-time product availability
- Tracking payments and pending dues
- Generating useful reports for decision-making

Table 3.1 System Functionalities According to Users

Sr. No.	Functionality	Admin	Staff	Customer
1	Login and authentication	Yes	Yes	Yes
2	View product catalog	Yes	Yes	Yes
3	Add or update categories	Yes	No	No
4	Add or update products	Yes	Limited	No
5	Manage customer records	Yes	Yes	Own profile
6	Create booking	Yes	Yes	Yes
7	Confirm or update booking status	Yes	Yes	No

Sr. No.	Functionality	Admin	Staff	Customer
8	Check product availability	Yes	Yes	Yes
9	Pre-issue inspection entry	Yes	Yes	No
10	Return and post-return inspection	Yes	Yes	No
11	Payment entry and settlement	Yes	Yes	No
12	View my bookings	No	No	Yes

3.1 Functional Scope

The functional scope of the proposed system covers the complete rental lifecycle, starting from product listing to booking closure and final settlement. It includes all major operations required to manage rental activities efficiently, ensuring accuracy, transparency, and improved customer service.

3.1.1 Product and Category Management

The system enables the administrator to manage products and categories in a structured and organized manner.

- Add, update, and deactivate product categories
- Maintain product details such as name, category, rental amount, and status
- Track real-time availability of each product
- Prevent invalid bookings by verifying product status before confirmation

In summary, this module ensures that the product catalog remains accurate, well-organized, and ready for efficient booking operations.

3.1.2 Booking Management

Booking management is a core functionality of the system, responsible for handling all reservation activities.

- Create bookings with customer details, rental dates, and duration
- Validate product availability before confirming bookings
- Track booking status such as pending, confirmed, active, returned, or cancelled

- Maintain booking history for future reference and reporting

In summary, this module minimizes booking conflicts and ensures systematic reservation management.

3.1.3 Customer Management

This module manages customer-related information required for rental transactions.

- Register and maintain customer profiles
- Link customers with their booking and payment history
- Enable quick customer search for repeat bookings
- Maintain contact details for communication and follow-up

In summary, customer information is centralized, making it easily accessible and efficient to use.

3.1.4 Inspection and Return Management

Inspection and return handling are essential for maintaining product quality and accountability.

- Record product condition before issuing
- Capture return date and condition after product return
- Record late returns, damages, or penalties
- Support final settlement based on product condition

In summary, this module ensures better control over rented assets and reduces disputes.

3.1.5 Payment Management

This module manages all financial transactions related to rentals.

- Record advance payments and pending amounts
- Update payment status throughout the booking lifecycle
- Capture penalties or additional charges for delays or damages
- Close bookings after final payment settlement

3.1.6 Reporting and Dashboard

The system provides various reports and dashboards for better decision-making and monitoring.

- Generate booking summary reports

- Provide product availability and utilization reports
- Track payments and pending dues
- Analyze return and inspection trends

In summary, this module helps management monitor system performance and make informed decisions.

3.2 In-Scope and Out-of-Scope

Table 3.2 Scope Boundary (In and out scope)

Sr. No.	Area	In Scope	Out of Scope
1	Product Handling	Product listing, availability tracking, category mapping	IoT-based physical tracking devices
2	Booking	Booking creation, update, cancellation, and status tracking	AI-based demand prediction
3	Returns	Return entry, inspection, damage and delay recording	Automated hardware-based inspection
4	Payments	Basic payment records, due tracking, and settlement	Full accounting ERP integration
5	Users	Role-based login and user management	Biometric attendance or HR modules
6	Reports	Operational and financial summary reports	Advanced BI and forecasting dashboards
7	Customer Side	Browse products, create and view own bookings	Marketplace-style multi-vendor platform

3.3 User Requirements

The system is designed to fulfil the requirements of different types of users, including Admin, Staff, and Customer. Each user has specific roles and responsibilities within the rental management system.

3.1.1 Admin

The Admin is responsible for overall control, configuration, and monitoring of the system.

- **User Management:** Add, update, or remove users and assign roles
- **Catalog Management:** Manage product categories and product details
- **Booking Supervision:** Monitor and control all booking activities
- **Payment Oversight:** Verify payments, settlements, and pending dues
- **Reporting:** Access complete reports across all system modules
- **System Control:** Ensure data accuracy and smooth system operation

In summary, the Admin requires full access and control to ensure efficient and reliable system management.

3.1.2 Staff

Staff members are responsible for handling day-to-day rental operations.

- **Booking Operations:** Create and update bookings efficiently
- **Availability Check:** Verify product availability before issuing
- **Inspection Handling:** Record product condition before issue and after return
- **Return Processing:** Complete return process and update booking status
- **Payment Entry:** Record received payments and pending balances
- **Basic Reports:** View operational reports as required

In summary, Staff requires fast and accurate operational features for smooth rental processing.

3.1.3 Customer

Customer requirements focus on ease of use, convenience, and transparency.

- **Easy Access:** Log in and browse available products
- **Booking Request:** Create bookings based on available items
- **Booking Visibility:** View booking status and history
- **Communication:** Receive updates related to booking, return, and payments
- **Profile Management:** Maintain and update personal details

In summary, Customers require a simple, transparent, and user-friendly experience.

3.2 Non-Functional Scope

The system also includes non-functional requirements to ensure quality,

performance, and usability.

- **Performance:** Fast response time for searching, booking, and payment processing
- **Reliability:** Consistent and accurate data across all modules
- **Security:** Role-based access control and secure authentication
- **Usability:** Simple and user-friendly interface for all users
- **Scalability:** Ability to handle growth in users, products, and bookings
- **Maintainability:** Modular design for easy updates and enhancements

3.3 Expected Outcomes of Project Scope

By implementing the proposed system, the following outcomes are expected:

- Improved control over rental inventory and product availability
- Reduction in manual errors during booking and return handling
- Faster and more accurate payment processing and settlement
- Enhanced customer service and transparency
- Accurate and reliable reports for better decision-making

This chapter explained the functional and non-functional scope of the system along with user requirements and expected outcomes. It defined the boundaries of the system and clarified the responsibilities of different users. The scope ensures that the system remains focused, efficient, and aligned with business requirements.

Chapter 4

Methodology

This chapter explains the methodology used to design and develop the Traditional Attire Rental Shop Management System. It describes the structured approach followed during development, including requirement analysis, system design, implementation, and testing. The project follows the Waterfall Model of SDLC to ensure systematic and controlled development.

The purpose of this chapter is to provide a clear understanding of how the system is planned and implemented step by step.

To keep development disciplined, the project follows the Waterfall Model of Software Development Life Cycle (SDLC), where each phase is completed and reviewed before moving to the next.

4.1 SDLC Approach Followed Waterfall Model

The Waterfall Model is selected because the project requirements are clear and can be implemented phase by phase. Each phase produces verified outputs before the next phase begins, ensuring systematic progress and better control.

Phases followed:

1. Requirement Analysis
2. System Design
3. Implementation
4. Testing
5. Deployment
6. Maintenance

This model provides a clear and structured approach to software development. Each phase is completed before moving to the next, ensuring better planning and execution. It helps in maintaining discipline throughout the project lifecycle. Proper documentation is maintained at every stage, which supports future improvements and maintenance. The model also makes it easier to track progress and manage project activities efficiently.

The following diagram shows the sequential phases of the Waterfall Model adopted for this project:

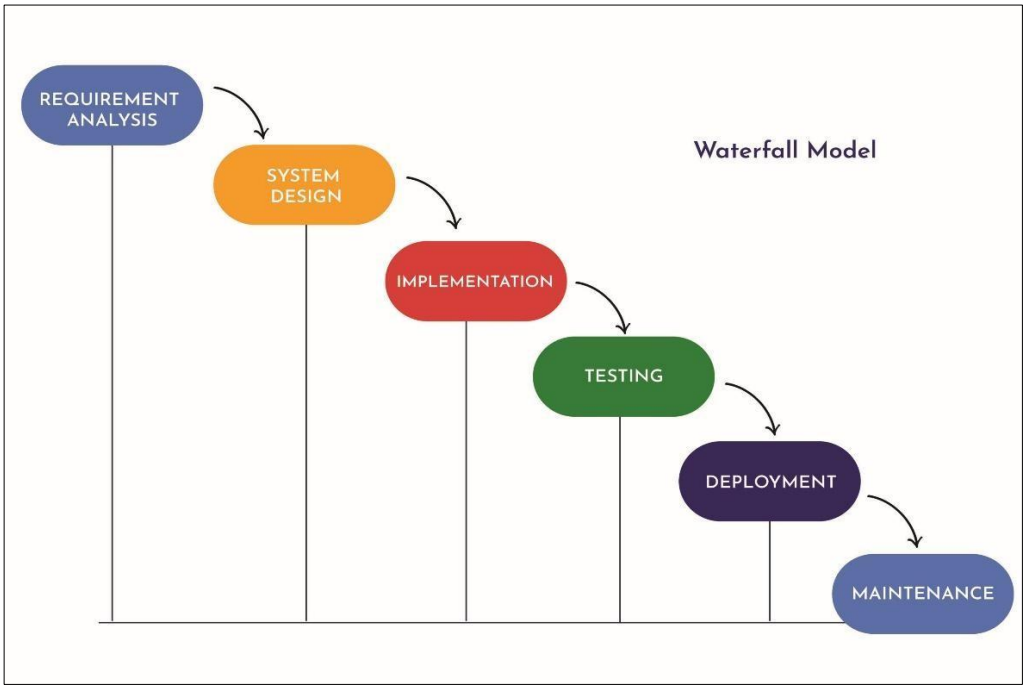


Figure 4.1: SDLC Waterfall Model — Sequential Phase Flow

1. **Requirement Analysis:-**In this phase, all the project requirements are collected and analyzed. The goal is to understand what the system should do. These requirements are clearly documented for future reference.

2. **System Design:-**Based on the requirements, the system architecture and design are created. This includes database design, system structure, and overall workflow of the application.

3. **Implementation:-**In this phase, the actual coding of the system takes place. Developers convert the design into a working application using programming languages and tools.

4. **Testing:-**The developed system is tested to identify and fix errors or bugs. This ensures that the system works correctly according to the requirements.

5. **Deployment:-**After successful testing, the system is deployed or released to users. It becomes available for real-world use.

6. **Maintenance:-**This is the final phase where the system is maintained and updated. Any issues found after deployment are fixed, and improvements are made as needed.

4.2 Requirement Analysis

Requirement analysis was performed through structured observation and user-level discussions to identify the actual needs of the rental business.

- Discussion with rental operations users
- Observation of manual booking and return workflows
- Study of common failures in manual systems
- Mapping required modules with user roles

Key Problems Identified in Existing / Manual Workflow

- Product availability is not updated in real time.
- Booking conflicts occur during peak demand.
- Return and inspection details are inconsistent.
- Payment due tracking is weak.
- Operational reports are delayed and error-prone.

4.2.1 Admin Requirements

The admin plays a crucial role in managing and controlling the overall system. Admin requirements focus on maintaining system operations, ensuring data accuracy, and monitoring all activities within the rental platform.

- Add and manage users with role assignments
- Maintain categories and products
- Monitor all bookings, returns, and payments
- Access dashboard and reports for decision-making
- Maintain system integrity and access control

4.2.2 Staff Requirements

Staff members are responsible for handling day-to-day operations of the rental system. Their requirements are focused on efficient execution of booking, return, and transaction- related activities.

- Create, update, and manage bookings
- Check product availability before booking confirmation
- Perform issue-time and return-time inspections
- Record and update payment details
- Complete return closure workflow.

4.2.3 Customer Requirements

Customers interact with the system to browse, book, and manage rental products. Their requirements emphasize ease of use, transparency, and quick access to booking and payment information.

- Browse products and check availability
- Create booking requests
- Track own bookings and status updates
- View payment summary for own transactions

4.3 System Design Methodology

The system design phase focuses on creating a structured plan for implementing the Traditional Attire Rental Shop Management System. This phase includes designing the system architecture, database structure, and user interface based on the requirements gathered from different users such as Admin, Staff, and Customers.

In this phase, the identified requirements are transformed into a detailed system design that defines the structure, workflow, and functionalities of the system. The system ensures smooth management of bookings, payments, returns, and inventory tracking.

Additionally, the system design helps in improving overall efficiency by reducing manual errors and ensuring accurate data handling. It also supports real-time updates of product availability and booking status, which enhances user experience. The system is designed to be scalable so that new features and modules can be added in the future. Proper validation and security mechanisms are also considered to protect user data and ensure reliable system performance.

Based on the identified requirements, the core functionalities of the system are defined as follows:

- Customer Management
- Booking Management
- Payment & Deposit Handling
- Inspection & Return Management
- Inventory Monitoring
- User Management

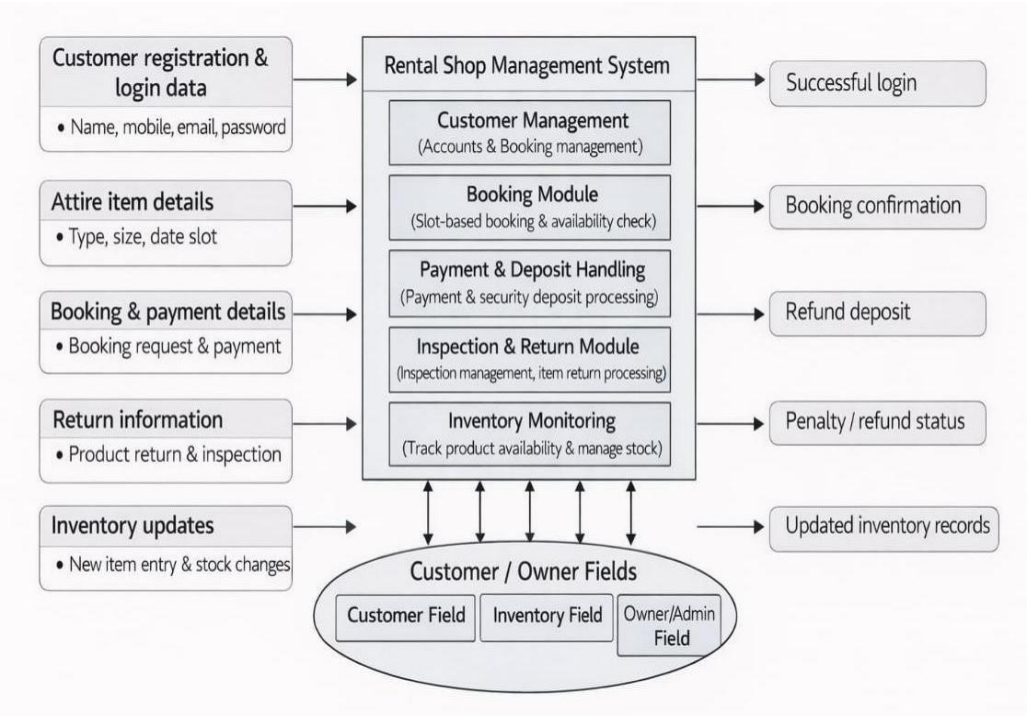


Figure 4.2: Proposed System for Traditional Attire Rental Shop Management System

Fig. 4.2 clarifies how user inputs such as customer details, booking information, payment data, return details, and product availability are processed by the system. The system interacts with the database and different modules to generate meaningful outputs that improve booking management, ensure accurate inventory tracking, and streamline rental operations.

4.3.1 Customer Management

In this section, we will explain how the system manages customer details, registration, and login activities.

- Stores important customer details such as name, contact number, email, and password.
- Allows customers to register and login into the system securely.
- Maintains customer booking history and account information.

In short, the system manages customer data efficiently and provides secure access to rental services.

4.3.2 Booking Management

In this section, we will describe how the system handles booking of rental items, including availability checking and confirmation.

- Allows customers to select products and rental dates.
- Checks product availability to avoid double booking.
- Records and confirms booking details in the system.

In short, this module ensures smooth and conflict-free booking of rental items.

4.3.3 Payment & Deposit Handling

In this section, we will cover how the system manages payment transactions and security deposit handling.

- Calculates rent based on rental duration.
- Records payment and deposit details accurately.
- Handles refund of deposit after product return.

In short, the system ensures transparent and accurate payment processing.

4.3.4 Inspection & Return Management

In this section, we will explain how the system manages product return and inspection process.

- Records return details such as return date and product condition.
- Checks whether the product is damaged or in good condition.
- Applies penalty if damage or delay is found.

In short, the system ensures proper return handling and maintains product condition records.

4.3.5 Inventory Monitoring

In this section, we will describe how the system tracks product availability and updates inventory.

- Tracks availability of rental products in real-time.
- Updates inventory after booking and return.
- Maintains accurate stock records for all items.

In short, the system helps maintain updated inventory and prevents booking conflicts.

4.4 User Management

In this section, we will cover how the system manages user access and ensures secure login for different roles such as Admin, Staff, and Customer.

- The system provides **role-based access control** for different users like Admin, Staff, and Customers.

- It ensures **secure authentication** using login credentials such as username and password. Each user can access only the features permitted to their role.

The proposed system diagram, shown in Fig. 4.2, illustrates the flow of data and interactions within our **"Traditional Attire Rental Shop Management System"**. The diagram is divided into the following main components:

i. Input:

- User registration details (Name, Mobile, Email, Password)
- Login credentials (Username & Password)
- User role selection (Admin / Customer)

ii. System:

- Validates user input (correct format, required fields)
- Authenticates login credentials
- Allows user registration, login, logout

iii. Database:

- Stores user details (User ID, Name, Email, Password)
- Stores login credentials securely
- Maintains user roles (Admin / Customer)
- Tracks user activity/history

iv. Output:

- Successful registration confirmation
- Login success / failure message
- Updated user profile information

4.5 Technology use to develop project:

- **Programming Language(s):** JavaScript
- **Frontend:** HTML, CSS, Bootstrap, React.js (using Vite.js)
- **Backend:** Node.js, Express.js
- **Database:** MongoDB
- **Payment Integration:** Razorpay (for online payment processing)
- **Development Tools:** GitHub (Version Control), Postman (API Testing), Visual Studio Code (Development)
- **Operating System:** Windows 11

This chapter described the development methodology, including requirement analysis, system design, and implementation approach. It explained how different modules such as booking, payment, inspection, and inventory are designed and integrated into the system. The methodology ensures that the system is reliable, scalable, and efficient.

Chapter 5

Details of Designs, Working and Processes

This chapter provides a detailed explanation of the system design, workflow, and operational processes of the Rental Management System. It describes how different modules interact with each other and how data flows through the system. The chapter includes system flow diagrams, module-wise operations, database design, and end-to-end working process.

The purpose of this chapter is to give a complete understanding of how the system functions internally.

5.1 System Flow Diagram

The Rental Management System manages complete rental operations from product setup to booking closure. It reduces manual errors by keeping all operational data in one centralized system and by updating records in real time. Every action performed by any user is immediately reflected across all connected modules, ensuring consistency and traceability throughout the rental lifecycle.

The system serves three user-facing roles Admin, Staff, and Customer each interacting with the system in a defined and controlled manner. All data operations are handled through a database-backed system that stores and retrieves information for every transaction, inspection, payment, and report.

The system has three user-facing roles and one central control role:

- Admin — system configuration and monitoring
- Staff — daily rental operations (booking, inspection, return, payment)
- Customer — browsing, booking requests, and status tracking
- Database-backed system services — real-time data storage and retrieval

The diagram below shows how each actor interacts with the central Rental Management System and what outputs are produced:

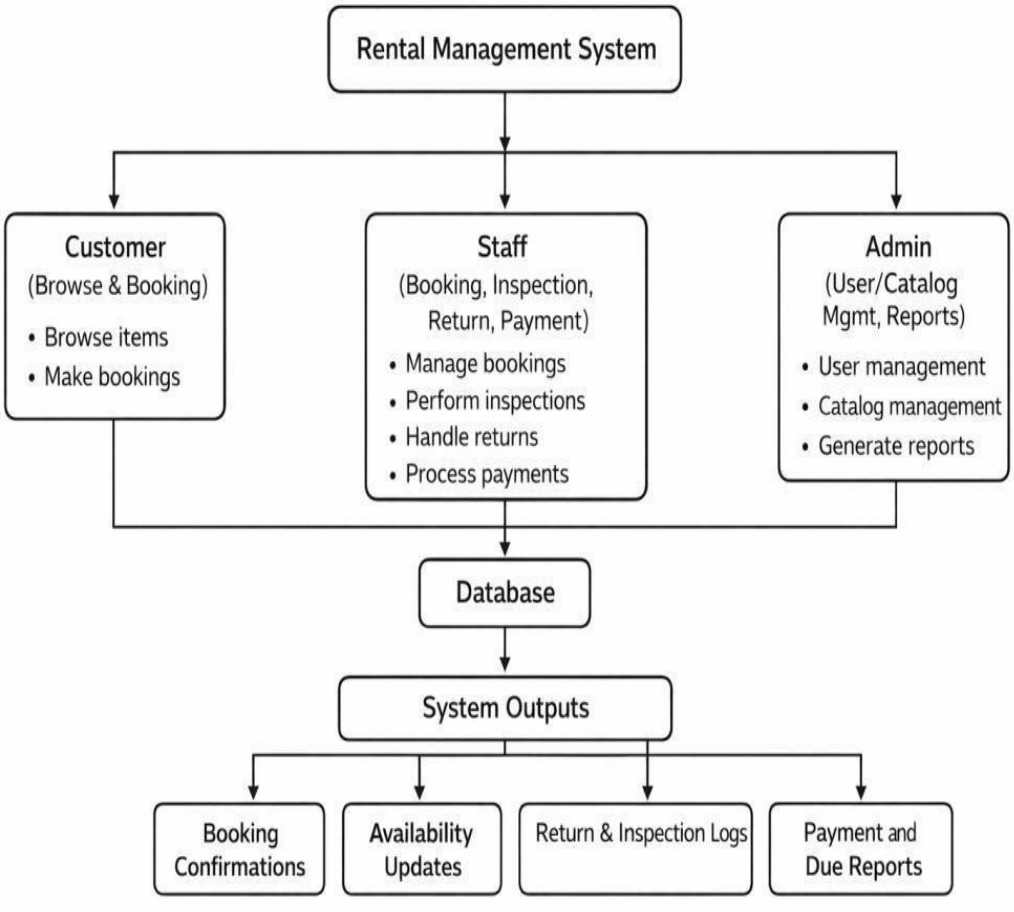


Figure 5.1: System Flow Diagram — Actors, System Core, and Outputs

1. Rental Management System (Top Level):- This is the main system that manages all rental-related activities. It connects customers, staff, and admin to perform different operations.

2. Customer Module (Browse & Booking):- Customers interact with the system to:

- Browse available items
- Make bookings

3. Staff Module (Booking, Inspection, Return, Payment):-Staff members handle operational tasks such as:

- Managing bookings
- Performing item inspections
- Handling returns

- Processing payments

4. Admin Module (User/Catalog Management, Reports):- The admin controls the overall system by:

- Managing users
- Managing product catalog
- Generating reports

  This module is responsible for monitoring and controlling system activities.

5. Database:-All data from customers, staff, and admin is stored in the database. It acts as a central storage for bookings, users, payments, and item details.

6. System Outputs:- The system generates useful outputs such as:

- Booking confirmations
- Availability updates
- Return and inspection logs
- Payment and due reports

5.2 Admin Module

The Admin Module is the control center of the application. It is responsible for configuring system masters and monitoring all operational activities across the platform. The Admin has complete visibility into all bookings, payments, returns, and user activities. This module ensures that the system remains consistent, secure, and aligned with business requirements.

Admin actions typically include setting up the rental catalog, assigning user roles, and reviewing operational reports to make informed business decisions. Without a properly functioning Admin Module, the rest of the system cannot operate effectively.

Key features:

- Manage users and role permissions.
- Add and update categories and products.
- Monitor bookings, returns, inspections, and payments.
- View dashboard and consolidated reports

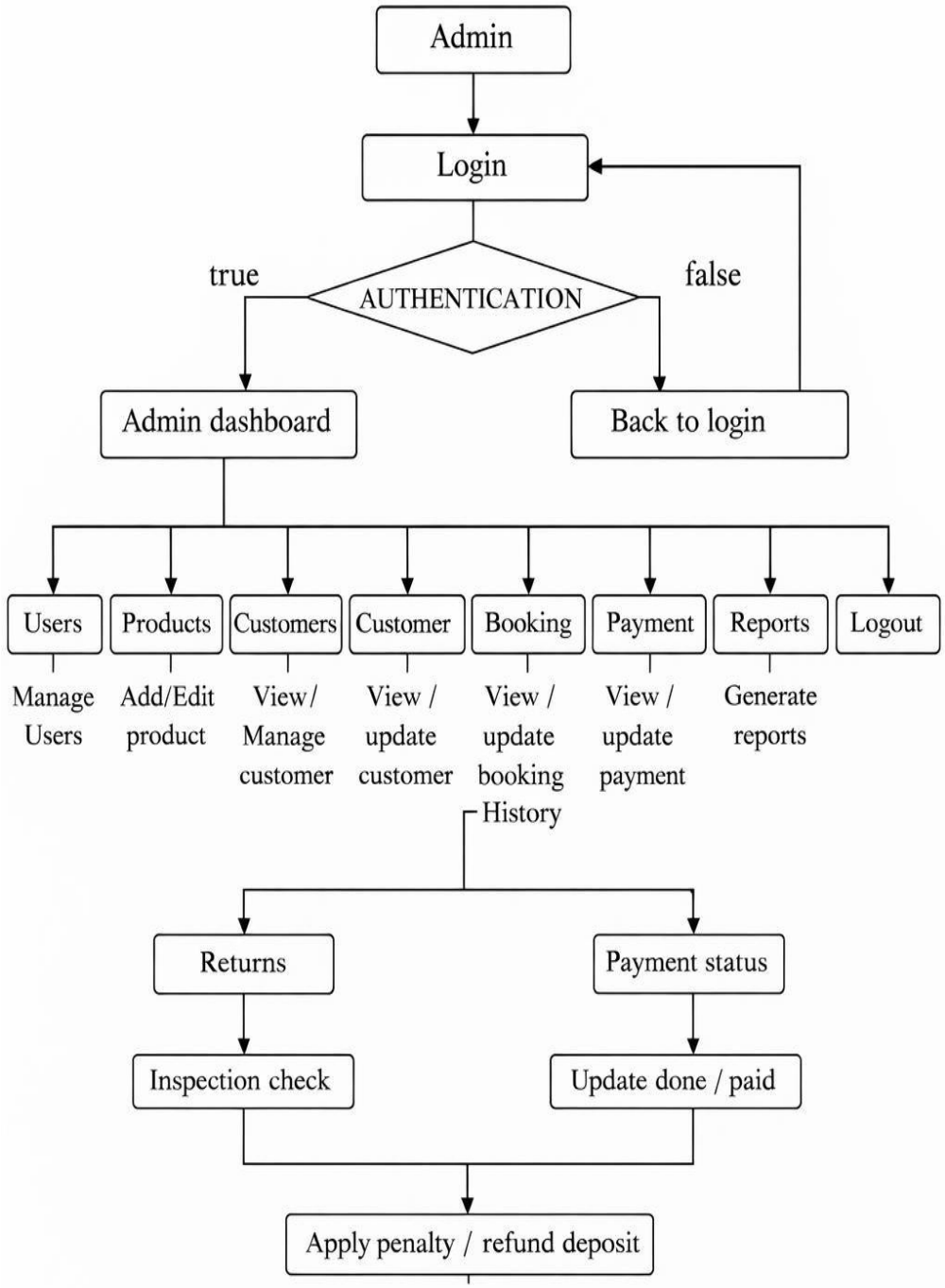


Figure 5.2: Admin Module Flow — Login to Report Generation and Logout

The diagram represents the workflow of the **Admin module** in the Rental Management System. It begins with admin login and authentication, after which the admin accesses the dashboard to manage system operations such as users, products, customers, bookings, payments, and reports. It also includes handling returns, inspection, and final actions like penalty or refund, ensuring smooth and controlled system management.

Key Points:

- Admin logs in and undergoes authentication
- On success, access to Admin Dashboard; otherwise back to login
- Manage users, products, and customers
- View and update booking history
- Monitor and update payment details
- Generate system reports
- Handle returns and perform inspection checks
- Update payment status (paid/pending)
- Apply penalty or refund deposit based on condition
- Logout securely from the system

5.3 Staff Module

The Staff Module handles all day-to-day operational workflows of the rental business. Staff members are the primary users who interact with customers, manage physical product handovers, and ensure that all rental transactions are properly recorded in the system. This module covers the entire lifecycle of a rental — from initial availability check to final settlement and booking closure.

Staff members do not have system configuration access, but they have full operational access to booking, inspection, return, and payment functions. The module is designed to be fast and efficient, minimizing the time required to complete each rental transaction.

Key features:

- Check product availability.
- Create and update bookings.
- Perform issue-time and return-time inspections.
- Process returns and update product status.
- Record advance and final payments.

The following flow diagram shows the complete Staff Module workflow including the damage/delay decision branch:

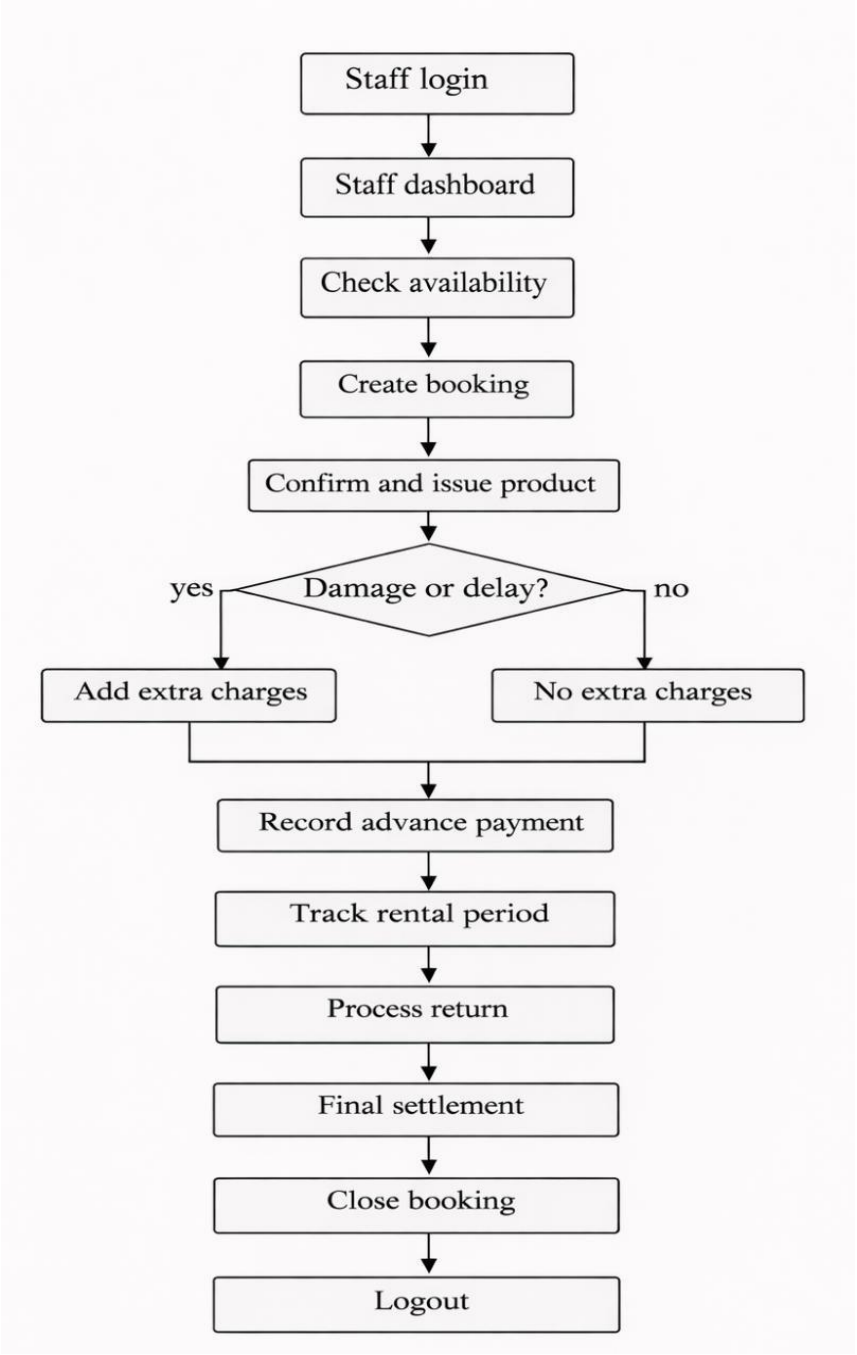


Figure 5.3 : Staff Module Flow

The diagram represents the workflow of the **Staff module** in the Rental Management System. It shows how staff members handle the complete rental process, from login to booking, issuing products, managing returns, and closing the transaction efficiently.

5.4 Customer Module

The Customer Module provides a simplified and transparent interface for customers to access rental services. Customers can browse the product

catalog, check availability for specific dates, submit booking requests, and track the status of their bookings. The module is designed to require minimal effort from the customer while keeping them informed at every stage of the rental process.

Customers do not have access to internal operational data such as inspection records or payment entries made by Staff. Their view is limited to their own bookings and associated payment details, which ensures data privacy and a focused user experience.

Key features:

- Browse products and check availability.
- Create booking request.
- Track own booking status.
- View payment details and due amount.

The following diagram represents the Customer Module workflow from login through booking and status tracking to logout:

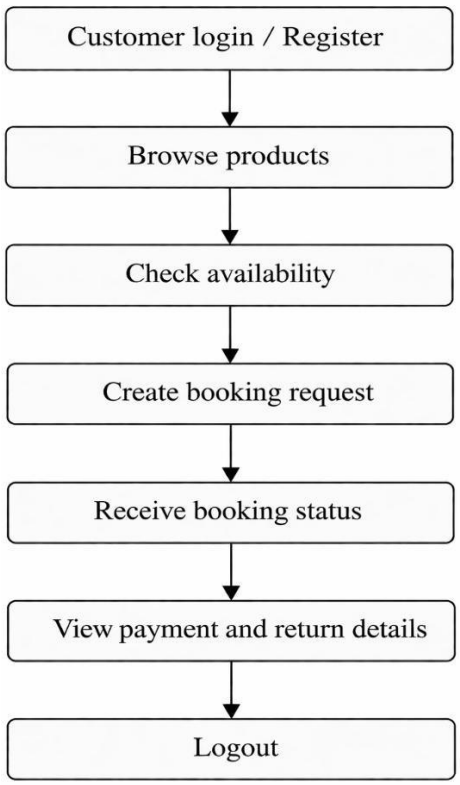


Figure 5.4 : Customer Module Flow

The diagram represents the workflow of the **Customer module** in the Rental Management System. It shows how a customer interacts with the system from login/registration to browsing products, making bookings, and viewing payment and return details.

Key Points:

- Customer registers or logs into the system
- Browses available products in the catalog
- Checks product availability before booking
- Creates a booking request for selected items
- Receives booking confirmation or status update
- Views payment details and return information
- Logs out after completing activities

5.5 Use Case Diagram

The Use Case Diagram identifies the system actors and maps them to the set of actions they are permitted to perform within the Rental Management System. It serves as a high- level view of system functionality from the perspective of each user role.

The system has three major actors — Admin, Staff, and Customer. Each actor interacts with a distinct set of use cases based on their role. Admin has the widest access, covering configuration and monitoring. Staff focuses on operational transactions. Customer has access only to product browsing and their own booking records.

Before using the system, different users (Customer, Staff, Admin) have specific roles and interactions with the Rental Management System:

- **Customer** interacts to browse products, create booking requests, and view bookings
- **Staff** helps in managing bookings and updating their status
- **Admin** controls the system, manages users, and monitors reports
- Core system functions include managing users, booking updates, inspection, return processing, and payment handling

This diagram is essential for understanding the boundary of each role and for ensuring that access control is correctly implemented in the system.

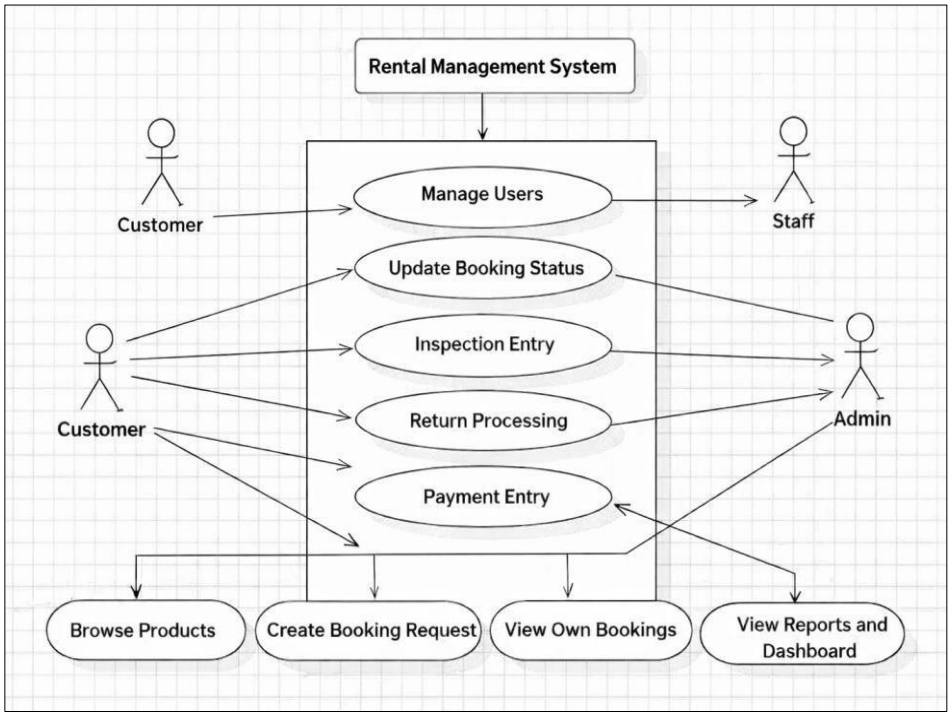


Figure 5.5: Use Case Diagram — System Actors and Role-wise Use Cases

After the user interacts with the system, the following operations are performed:

- Customers can successfully browse products and place booking requests
- Booking status is updated and visible to users
- Staff and admin perform inspection and return processing
- Payments are recorded and managed properly
- Admin can view reports and overall system dashboard
- The system ensures smooth coordination between all modules and users

5.6 Database Schema

The database is the backbone of the Rental Management System. It stores all operational data in a structured format and supports the complete rental lifecycle — from user authentication and product cataloging to

booking management, inspection records, return tracking, and payment settlement.

The schema is designed around eight core entities, each representing a distinct operational concept. Relationships between entities are carefully defined to ensure that data integrity is maintained across all operations. For example, a booking record is always linked to a specific customer, product, and user, ensuring full traceability for any transaction.

The database structure also supports reporting by allowing cross entity queries. Dashboard summaries and operational reports are generated by joining booking, payment, return, and inspection data.

The following table describes all database entities, their purpose, and the key fields they contain:

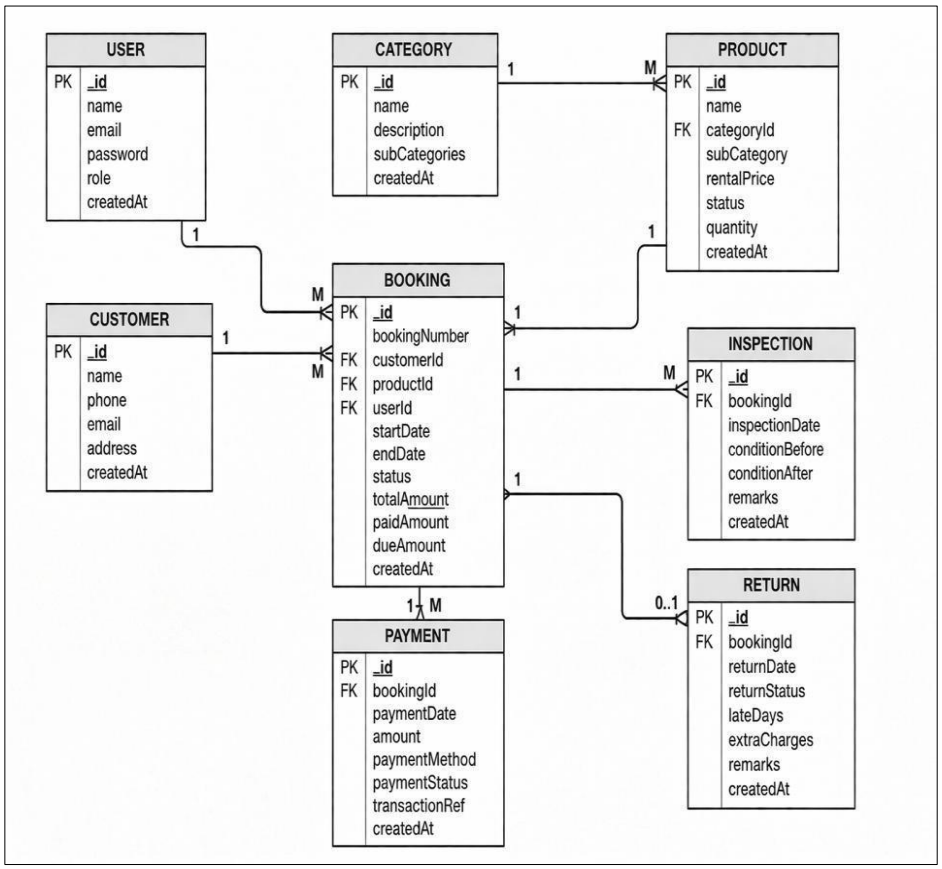


Figure 5.6 : Database ER Diagram — Entity Descriptions and Key Fields Entity Relationships

The ER diagram represents the structured database design of the Rental Management System, where all entities and their relationships are defined before system execution. It includes tables such as User, Customer, Category, Product, Booking, Payment, Inspection, and Return, with

Booking acting as the central entity connecting all modules. These relationships ensure proper data organization and integrity. After system execution, this structure supports smooth data flow, where bookings are created, products are managed, inspections are recorded, payments are processed, and returns are handled. The system maintains accurate records of transactions, due amounts, and reports, enabling efficient and reliable rental management.

The following table describes the relationships between the core database entities:

Table 5.1: Entity Relationship Description

From Entity	To Entity	Relationship Description
USER	BOOKING	One user (Admin/Staff) can create or manage many bookings.
CUSTOMER	BOOKING	One customer can place many bookings over time.
CATEGORY	PRODUCT	One category can contain many products.
PRODUCT	BOOKING	One product can appear in many bookings (across time periods).
BOOKING	INSPECTION	Each booking has at most one inspection record.
BOOKING	RETURN	Each booking has at most one return record.
BOOKING	PAYMENT	Each booking can have multiple payment entries.

List of Database Tables / Collections:-

The system uses the following eight database collections. Each collection is focused on a specific operational area and is linked to related collections through reference fields:

Table 5.2 : Database Collections — Names, Descriptions, and Key Fields

Collection Name	Description	Key Fields
users	Stores system users for authentication and role access.	_id, name, email, password, role, createdAt
categories	Stores product categories and optional subcategories.	_id, name, description, subCategories, createdAt
products	Stores rentable item details and category links.	_id, name, categoryId, subCategory, rentalPrice, status, quantity
customers	Stores customer identity and contact details.	_id, name, phone, email, address
bookings	Central transaction table for rental bookings.	_id, bookingNumber, customerId, productId, userId, status, totalAmount, dueAmount
inspections	Stores item condition records for issue/return checks.	_id, bookingId, inspectionDate, conditionBefore, conditionAfter, remarks
returns	Stores return completion details and extra charges.	_id, bookingId, returnDate, returnStatus, lateDays, extraCharges
payments	Stores payment transaction and settlement details.	_id, bookingId, paymentDate, amount, paymentMethod, paymentStatus

5.7 Working Process (End-to-End)

The end-to-end working process of the Rental Management System describes how a rental transaction progresses from initial admin setup all the way through to final booking closure and report generation. This process ties together all modules and roles in a single coherent operational flow.

Each step in the process produces a specific output that feeds into the next step. This ensures that no step can be skipped and that all required data is captured before the process moves forward. The system enforces this discipline through status-based controls and validation checks at each stage.

The complete working process follows these steps:

1. Admin configures users, categories, and products.
2. Staff verifies availability and creates booking.
3. Product is issued after confirmation and optional advance payment.
4. During or after the rental period, return is recorded.
5. Inspection is completed and damage or delay is computed.
6. Payment settlement is updated with any extra charges.
7. Booking is closed and reports are updated.

The design and process methodology of the Rental Management System ensures that all operations are connected through a consistent and traceable workflow. Each module from user management and catalog setup to booking lifecycle, inspection, return, and payment is clearly defined and linked to the others through the database schema and role-based access controls.

5.8 Payment Integration Flow

The payment workflow is initiated when the customer selects the “Pay Now” option on the frontend interface. This triggers a request to the backend server, where a payment order is generated. The backend then communicates with the Razorpay payment gateway to create an order and obtain a unique Order ID. This Order ID is returned to the frontend, which uses it to launch the Razorpay checkout interface.

The customer proceeds by entering the required payment details, and Razorpay processes the transaction securely. Upon completion, a payment response is generated and sent to the frontend, which forwards it to the backend for verification. The backend validates the transaction to ensure its authenticity

and integrity. Once verified, the payment status is updated in the database, and a confirmation response is sent back to the frontend, which displays a successful payment message to the customer.

- Customer initiates payment by clicking “Pay Now” on the frontend
- Frontend sends a payment request to the backend server
- Backend creates a payment order
- Backend interacts with Razorpay API to generate an Order ID
- Razorpay returns the Order ID to the backend
- Backend forwards order details to the frontend
- Frontend initiates Razorpay Checkout
- Customer provides payment credentials (Card/UPI/Net Banking)
- Razorpay processes the payment transaction
- Payment response is sent to the frontend
- Frontend forwards response to backend for verification
- Backend performs payment validation and verification
- Backend updates payment status in the database
- Frontend displays payment confirmation (success/failure) to the customer

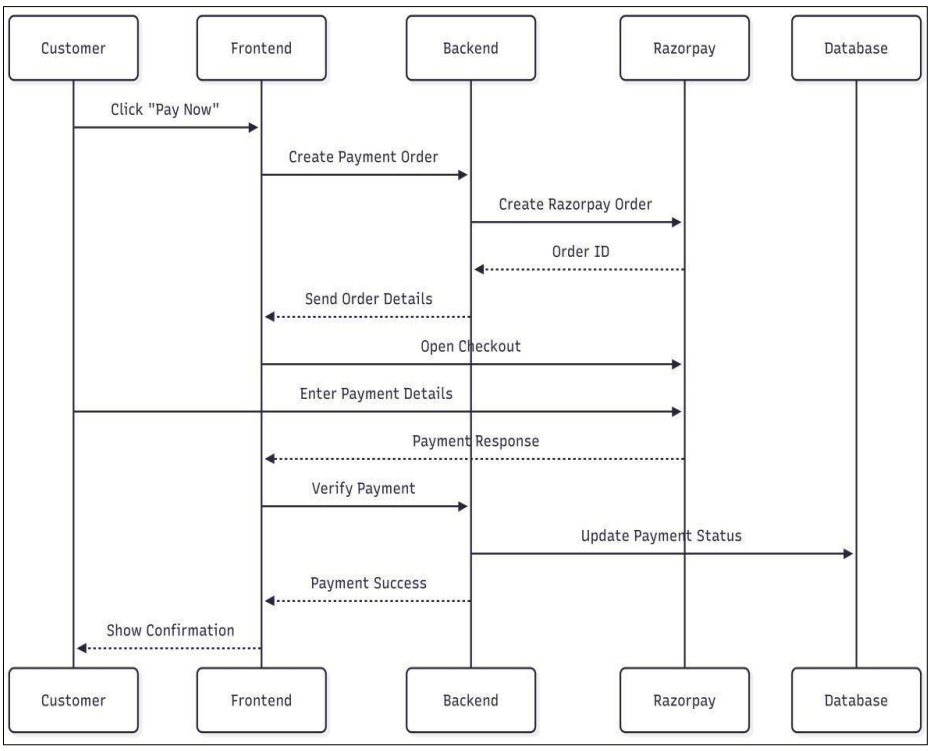


Figure 5.8 : Payment Integration Flow

This chapter explained the complete system design, including flow diagrams, module operations, database schema, and end-to-end working process. It demonstrated how different components of the system are interconnected and how data is managed efficiently. The structured design ensures accuracy, transparency, and smooth operation of the rental system.

Chapter 6

Result and Application

This chapter presents the results of the developed Rental Management System through major web pages and their functional descriptions. The screenshots demonstrate how different users interact with the system for rental operations such as product browsing, booking, return handling, payment, and report monitoring.

6.1 Website Sitemap

The following sitemap represents the structure of the Rental Management System web application. It shows the hierarchical arrangement of all major web pages and how users can navigate through different sections of the system. The sitemap is divided into public pages, customer pages, and admin pages, providing a clear overview of system navigation and functionality.

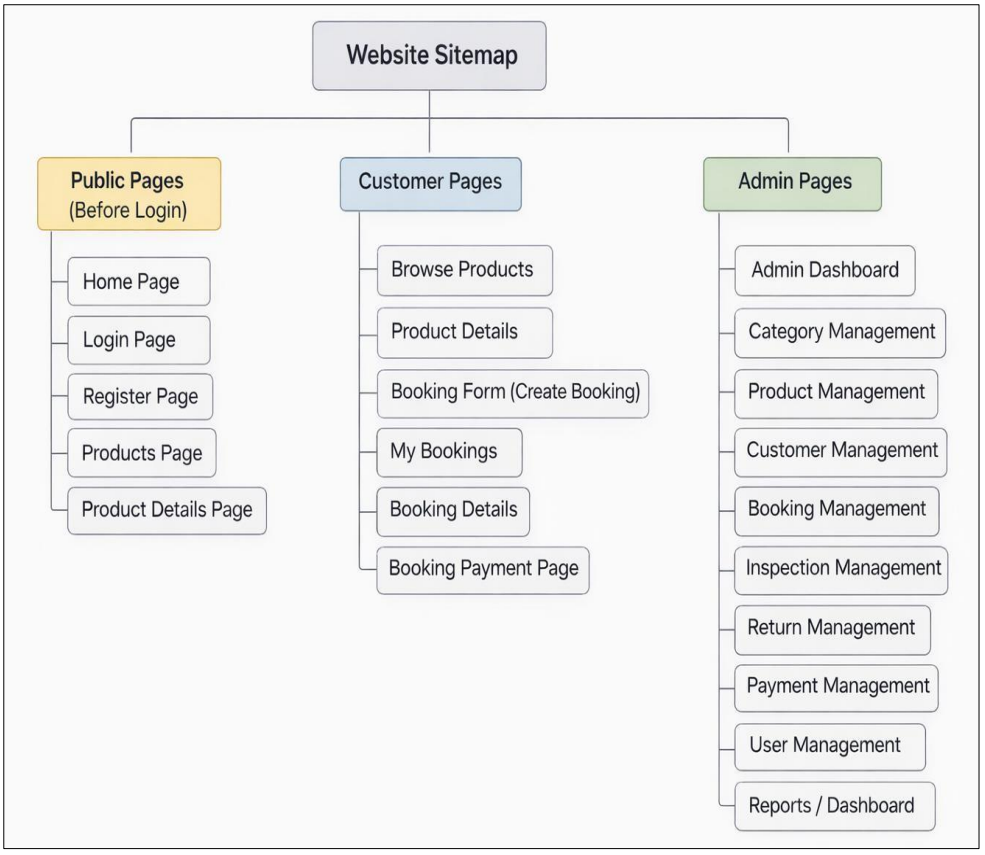


Figure 6.1: Website Sitemap of Rental Management System

6.2 Home Page

Figure 6.2 shows the home page of the Rental Management System. It introduces the platform and provides navigation to key sections such as products, about, contact, login, and registration.

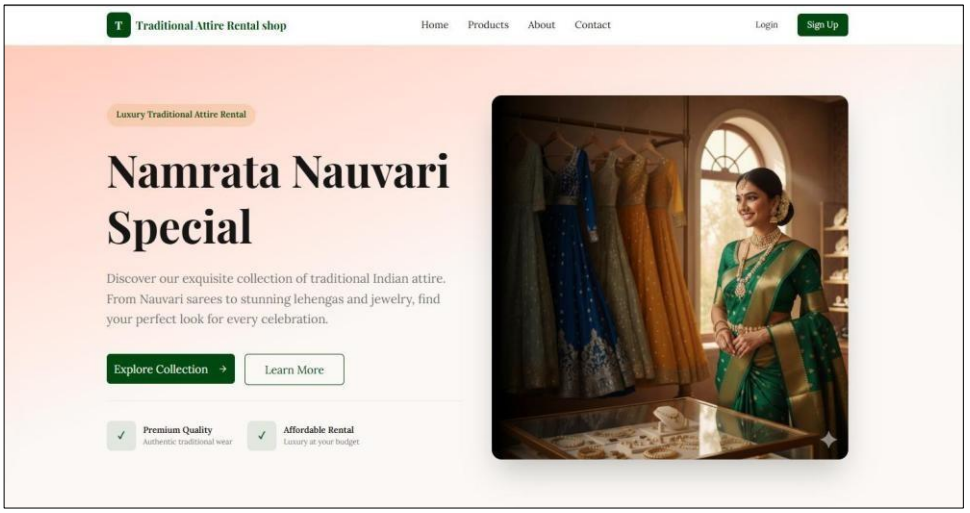


Figure 6.2: Home Page

Main outcomes visible on this page:

- **User-friendly Interface:** The home page provides a welcoming and clean layout for first-time visitors, reducing the time needed to understand the platform.
- **Clear Call-to-Action:** Login and registration links are prominently placed so users can quickly access the system.
- **Product Discovery:** Featured or listed rental items are visible on the home page, giving visitors an early look at available products Login Page

6.3 Login Page

shows the login page where registered users can access the system based on their role. The login form captures email and password credentials and authenticates the user against stored records. Upon successful login, the system redirects the user to their role-specific dashboard — Admin goes to the admin panel, Staff goes to the operational dashboard, and Customer goes to the product browsing view.

The login page also includes input validation to ensure that empty submissions are rejected, and it displays appropriate error messages when credentials do not match. This prevents unauthorized users from

proceeding further into the application.

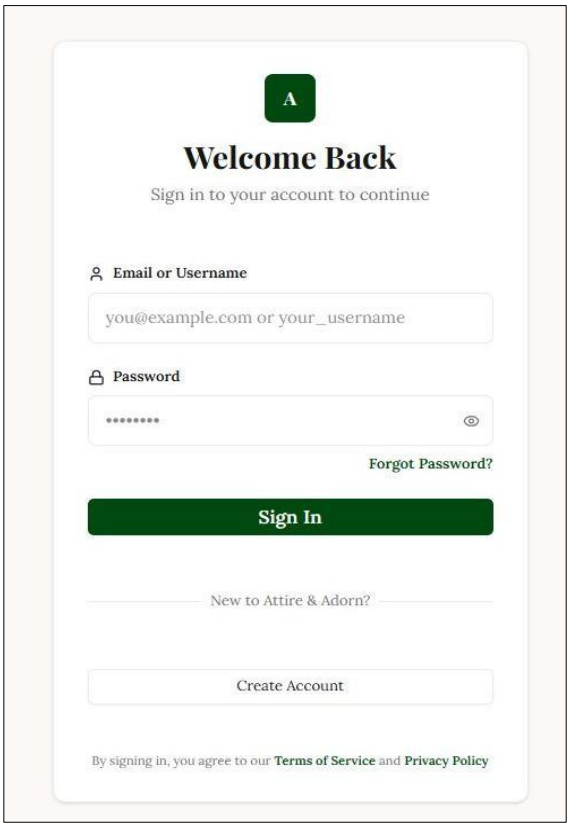


Figure 6.3: Login Page

Table 6.1: Login Validations – Secure Credential-Based Authentication

Fields	Validation Applied
Email / Username	"^[A-Za-z0-9_]{3,15}$" , "User must enter valid email format (e.g., user@gmail.com) OR username (3–15 characters, alphanumeric only)"
Password	"(?=.*\d)(?=.*[a-z])(?=.*[A-Z]).{8,16}", "Password must contain at least one number, one uppercase and one lowercase letter, and length between 8 to 16 characters"
Forgot Password	Must redirect to password recovery page, "User can reset password using registered email"

Fields	Validation Applied
Login Button	Enabled only when all required fields are valid, "Prevents submission of empty or invalid data"
Empty Fields Validation	"Email/Username and Password fields must not be empty"
Invalid Credentials	"Display error message if email/username or password is incorrect"
Account Lock (Security)	"After 3–5 failed attempts, account temporarily locked for security purpose"

6.4 Register Page

Figure 6.4 shows the registration page for new users. This page captures user details such as name, email, phone number, and password, then creates a new account in the system.

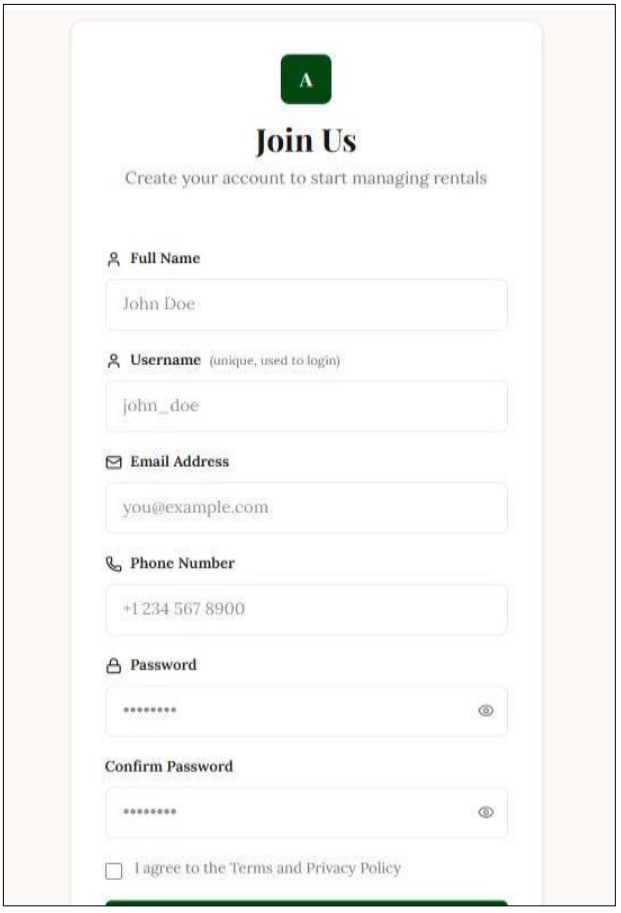


Figure 6.4: Register Page

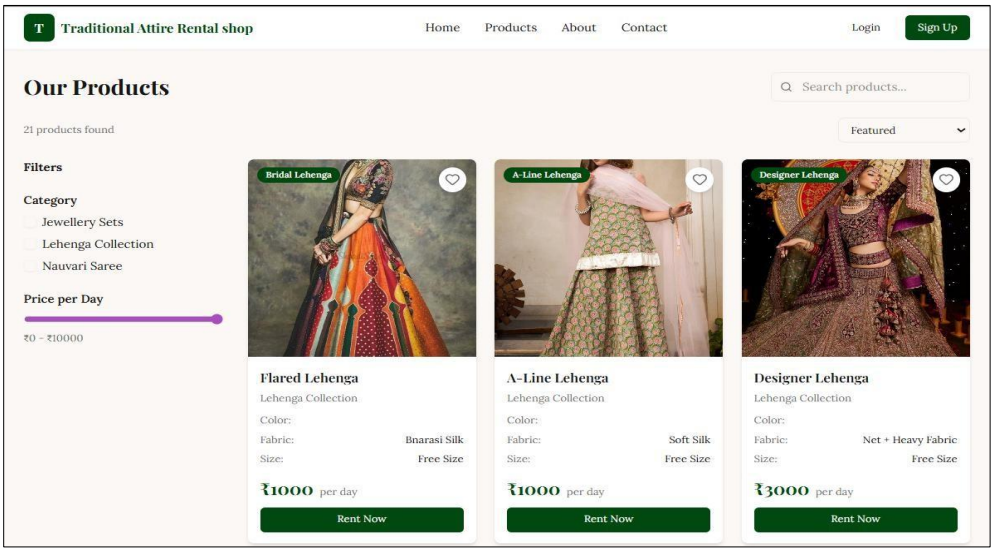
Table 6.2: Register Page Validations – User Account Creation

Fields	Validation Applied
Full Name	"^[A-Za-z]{3,30}$" , "Only alphabetic characters allowed, minimum 3 and maximum 30 characters"
Username	"^[A-Za-z0-9_]{3,15}$" , "Username must be 3–15 characters long and contain only letters, numbers, and underscore. Must be unique"
Email Address	"^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$" , "Must be a valid email format (e.g., user@gmail.com) and unique for each user"
Contact Number	"^[789][0-9]{9}$" , "Mobile number must be 10 digits and start with 7, 8, or 9"
Password	"(?=.*\d)(?=.*[a-z])(?=.*[A-Z]).{8,16}" , "Must contain at least one number, one uppercase and one lowercase letter, and be 8–16 characters long"
Confirm Password	Must match Password field, "Ensures user re-enters correct password"
Agree Terms & Policy	Checkbox (Required), "User must accept Terms & Conditions and Privacy Policy before registration"
Empty Fields Validation	"All fields are mandatory and must not be left blank"
Register Button	Enabled only when all validations are satisfied, "Prevents submission of invalid or incomplete data"

6.5 Products Page

Figure 6.5 shows the products listing page. Customers can view all rentable products available on the platform, along with key details such as product name, category, rent price per day or period, and current availability status. Products are organized by category and subcategory,

allowing customers to filter based on what they need.



.Figure 6.5: Products Page

Table 6.3: Product Page Validations – Product Browsing & Selection

Fields	Validation Applied
Category Selection	"User must select a category (e.g., Nauvari Saree, Accessories, etc.), default option should not be empty"
Search Product	"^[A-Za-z0-9]{1,30}$" , "Allows alphanumeric characters, minimum 1 and maximum 30 characters for search input"
Filter Options	"User can filter products based on size, price, color, or availability; at least one valid option should be selected if applied"
Sort By	"Products can be sorted by price (low to high / high to low) or popularity"
Product Quantity	"^[1-9][0-9]*$" , "Quantity must be a positive number and should not exceed available stock"
View Product Details	"User can view full product description, images, price, and availability"

Fields	Validation Applied
Out of Stock Handling	"If product is unavailable, 'Out of Stock' message should be displayed and add to cart disabled"
Empty Search Result	"Display message like 'No products found' if search does not match any item"
Price Validation	"Price must be displayed correctly and should be a positive value"

6.6 Product Details Page

Figure 6.6 shows the detailed information page for a selected product. When a customer clicks on a product from the listing page, they are taken to this page where all relevant details are displayed including a full description, rental price breakdown, deposit requirements, available quantity, category and subcategory classification, and any usage conditions or notes associated with the product.

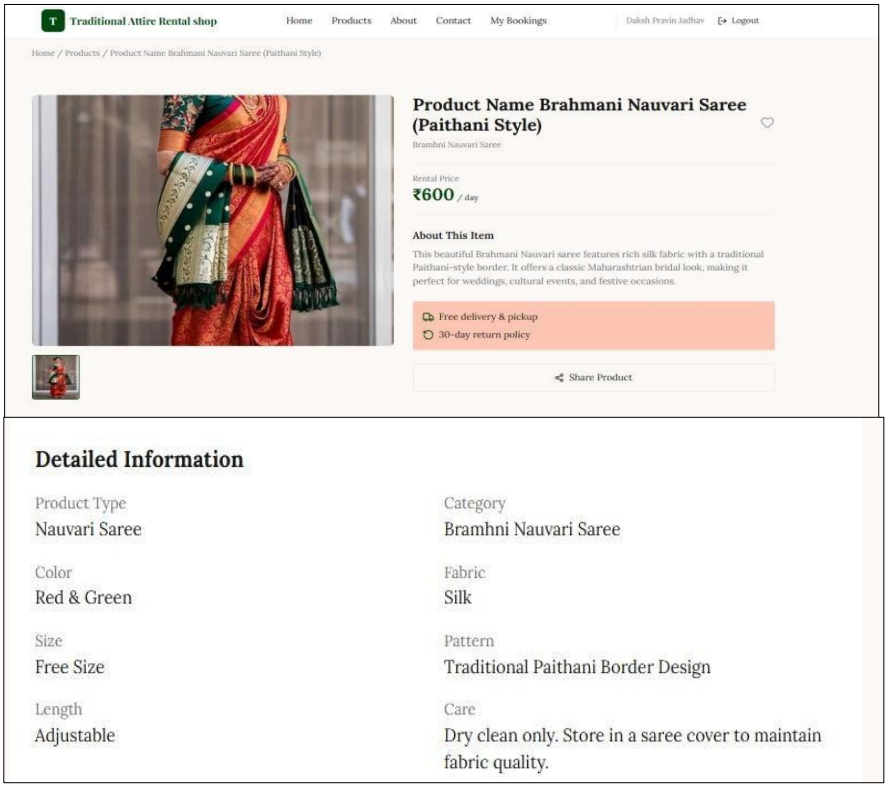


Figure 6.6: Product Details Page

Table 6.4: Product Details Page Validations – Product Information Display

Fields	Validation Applied
Product Name	"^[A-Za-z0-9]{3,50}$" , "Product name must be 3–50 characters long and contain valid text"
Price	"^[0-9]+(\.[0-9]{1,2})?$" , "Price must be a positive numeric value (can include decimal up to 2 digits)"
Product Type	"User must select or view valid product type (e.g., Nauvari Saree, Bridal Wear, Accessories)"
Color	"At least one color option must be available and selectable"
Size	"User must select a valid size (e.g., Free Size, S, M, L) before proceeding"
Length	"^[0-9]{1,3}$" , "Length must be a numeric value (in meters) within valid range"
Product Description	"Detailed description must be displayed; should not be empty"
Product Image	"Valid image must be displayed; placeholder shown if image is missing"
Availability Status	"Product must show availability (In Stock / Out of Stock)"
Add to Cart Button	Enabled only when required selections (size, color) are made and product is available
Error Handling	"Display message if required options (size/color) are not selected"

6.7 Booking Form / Booking Creation

Figure 6.7 shows the booking form where all necessary details for a rental are captured. The form includes customer information, the selected product, the rental start date, the rental end date, and the calculated duration. The system automatically computes the total rental amount based on the selected dates and the product's daily or periodic rate.

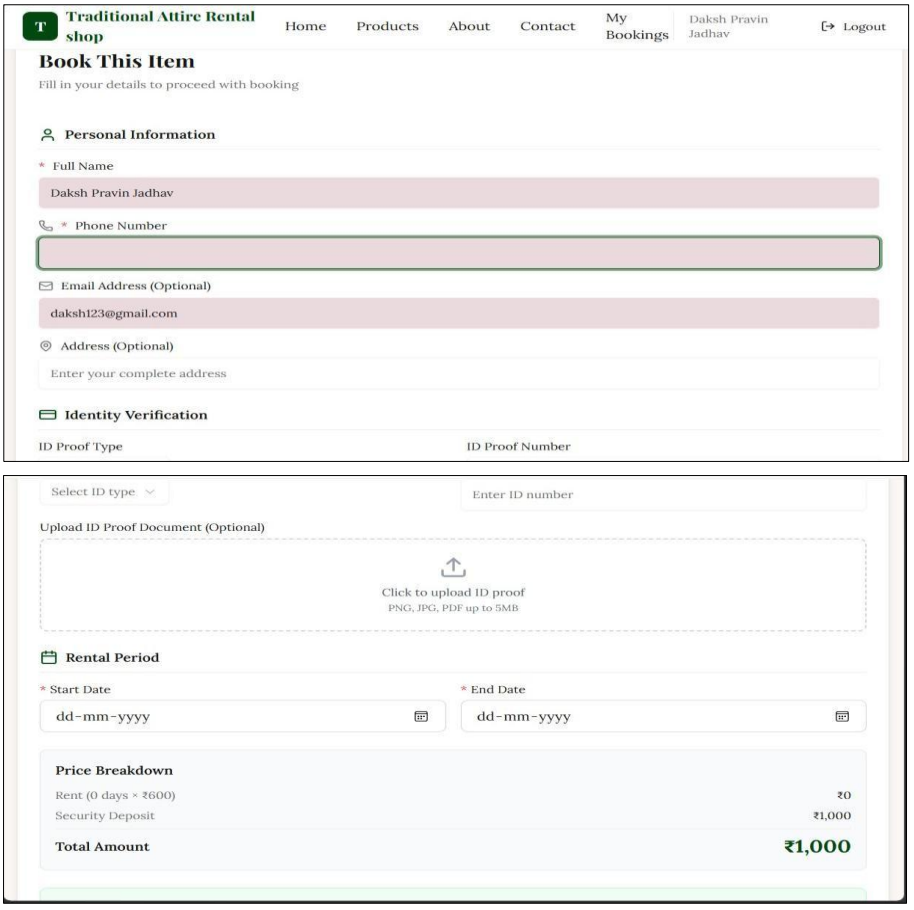


Figure 6.7: Booking Form

Table 6.5: Booking Form Page Validations – Rental Booking Process

Fields	Validation Applied
Full Name	"^[A-Za-z]{3,30}$" , "Only alphabetic characters allowed, minimum 3 and maximum 30 characters"

Fields	Validation Applied
Phone Number	"^[789][0-9]{9}$" , "Mobile number must be 10 digits and start with 7, 8, or 9"
Email Address	"^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$" , "Must be a valid email format (e.g., user@gmail.com)"
Address	"^.{10,100}$" , "Address must be between 10 to 100 characters and should not be empty"
ID Proof	"User must select a valid ID proof (e.g., Aadhaar Card, PAN Card, Driving License)"
Image Upload	`"(.jpg)
Start Date	Must be current or future date, "Booking start date cannot be in the past"
End Date	Must be greater than Start Date, "End date should be after start date"
Total Amount	"^[0-9]+(\.[0-9]{1,2})?$" , "Auto-calculated based on rental days; must be a positive value"
Date Validation	"System calculates total rental duration based on start and end date"
Submit/Book Button	Enabled only when all validations are satisfied

6.8 My Bookings Page

Figure 6.8 shows the booking history page for the logged-in customer. This page lists all past and current bookings associated with the customer's account. Each booking card or row displays the booking number, product name, rental dates, current status, and a link to view full booking details.

The page is designed to provide a clear and organized view of all bookings, making it easy for customers to track their rental activities. It also allows users to quickly identify active, completed, or pending bookings through status indicators. Additionally, customers can access detailed information such as payment status, return details, and any additional charges. This feature enhances transparency and improves user experience by keeping all booking-related information in one place.

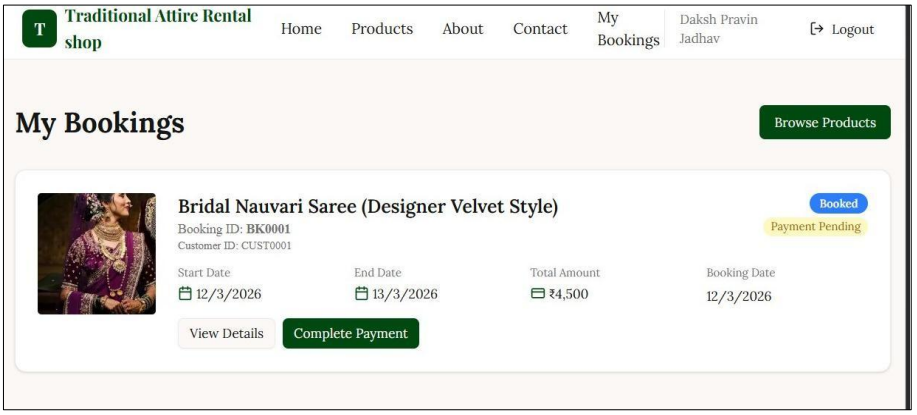


Figure 6.8: My Bookings Page

Table 6.6: My Booking Page Validations – User Booking Management

Fields	Validation Applied
Booking ID	"Each booking must have a unique ID generated by the system"
Product Name	"Displays booked product name correctly; should not be empty"
Product Image	"Valid product image must be displayed; placeholder if missing"
Booking Date	"Displays the date when booking was made (auto-generated)"
Start Date	"Must match the selected booking start date; cannot be modified"

Fields	Validation Applied
End Date	"Must match the selected booking end date; should be greater than start date"
Total Amount	"^[0-9]+(\.[0-9]{1,2})?$" , "Displays calculated rental amount; must be positive"
Booking Status	"Shows current status (Booked / Ongoing / Completed / Cancelled)"
Search Booking	"^[A-Za-z0-9]{1,30}$" , "Allows searching booking by product name or ID"
Filter Bookings	"User can filter bookings based on status or date"

6.9 Booking Details Page

Figure 6.9 shows the detailed booking information page. When a customer clicks on a specific booking from the My Bookings page, they are taken to this screen which shows the complete booking record — including product details, rental dates, current booking status, payment summary (paid, pending, total), return date if applicable, and any inspection notes or remarks that have been added by staff.

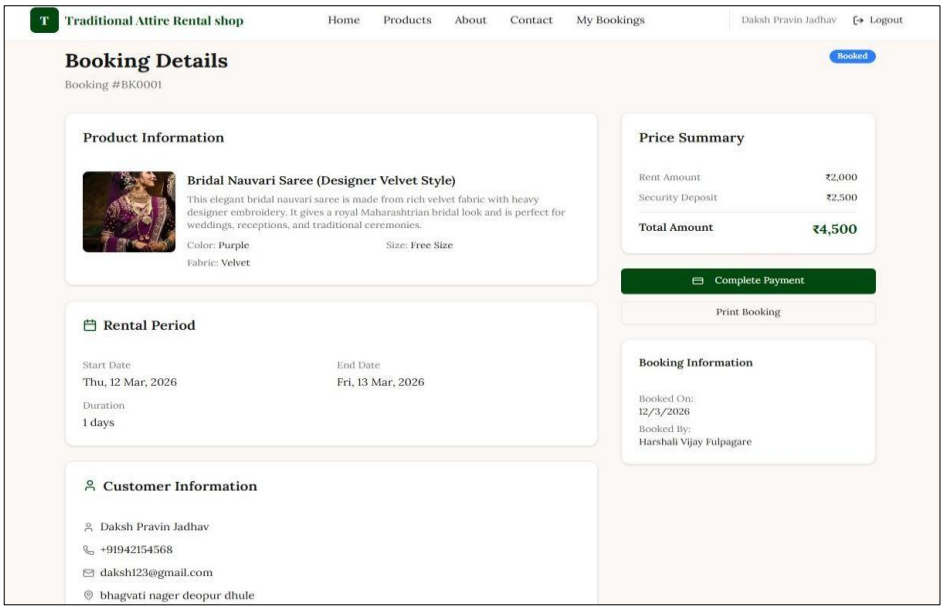


Figure 6.9: Booking Details Page

Table 6.7: Booking Details Page Validations – Detailed Booking Information

Fields	Validation Applied
Booking ID	"Unique ID generated by system; cannot be edited"
Customer Name	"Displays logged-in user name; read-only field"
Phone Number	"^[789][0-9]{9}$" , "Must be valid 10-digit mobile number"
Email Address	"^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$" , "Must be valid email format"
Address	"^.{10,100}$" , "Address must be between 10 to 100 characters"
Product Name	"Displays correct booked product; cannot be empty"
Product Type	"Shows selected category/type (e.g., Nauvari Saree, Bridal Wear)"
Product Image	"Valid image must be displayed; placeholder if missing"
Total Days	"Automatically calculated based on start and end date"
Booking Status	"Displays status (Booked / Ongoing / Completed / Cancelled)"
Payment Status	"Displays payment state (Paid / Pending / Failed)"
Cancel Booking Option	Enabled only before start date; restricted after booking begins

6.10 Booking Payment Page

Figure 6.10 shows the payment page associated with a specific rental transaction. This page displays the total rental amount, the advance paid at the time of booking, any additional charges added due to damage or delay, the remaining due amount, and the full payment history for the booking.

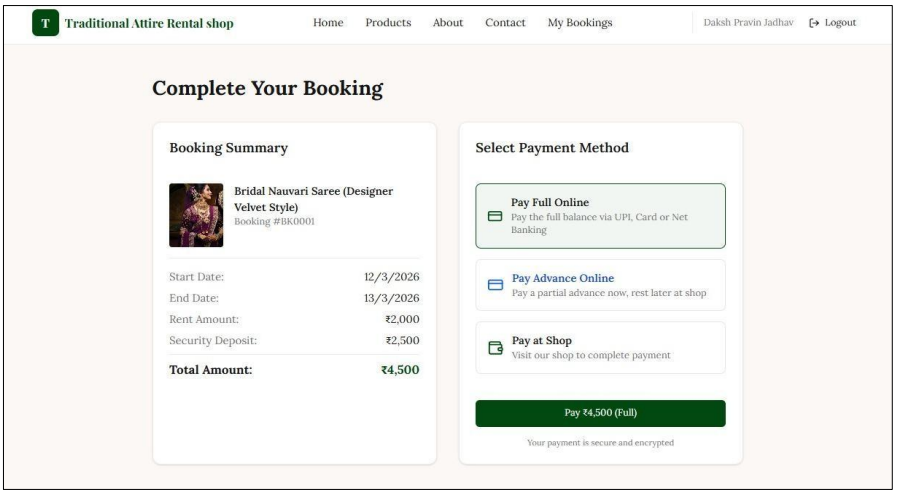


Fig. 6.10: Booking Payment Page

Table 6.8: Booking Payment Page Validations – Payment Processing

Fields	Validation Applied
Payment Option	"User must select one option: Pay Full Online / Pay Advance Online / Pay at Shop"
Pay Full Online	"User must pay total amount online; booking confirmed only after successful payment"
Pay Advance Online	"User must pay a predefined advance amount (e.g., 20–50% of total); remaining amount to be paid at shop"
Pay at Shop	"No online payment required; full payment will be done at the shop during pickup"

Fields	Validation Applied
Total Amount	"^[0-9]+(\.[0-9]{1,2})?$" , "Auto-calculated (Rent × Days); must be positive"
Payment Gateway Validation	"Secure payment gateway used for online transactions (UPI/Card/Net Banking)"
Payment Confirmation	"Booking confirmed only after successful payment (for online modes)"

6.11 Admin Dashboard

Figure 6.11 shows the Admin dashboard. It provides a centralized overview of major system metrics including total users, active products, pending and active bookings, pending returns,and payment summaries. The dashboard is the first page Admin sees after login and is designed to give a quick, data-driven snapshot of the current state of operations.

Each metric card or panel on the dashboard links directly to the detailed module it represents.

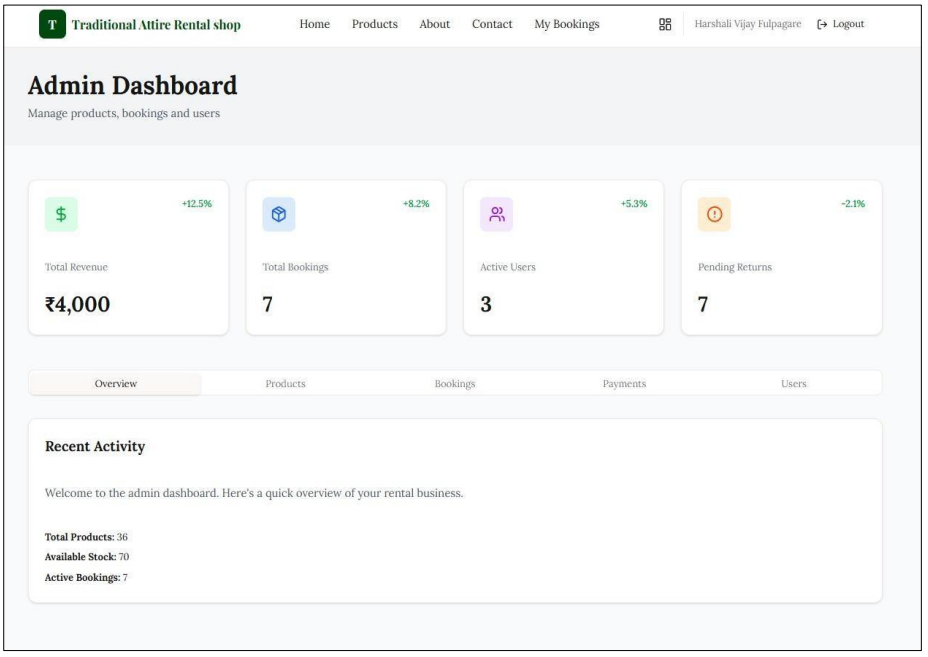


Figure 6.11: Admin Dashboard

Table 6.9: Admin Dashboard Validations – System Overview & Management

Fields	Validation Applied
Total Revenue	"^[0-9]+(\.[0-9]{1,2})?$" , "Displays total earnings from all completed bookings; must be positive and auto-calculated"
Total Bookings	"^[0-9]+$" , "Shows total number of bookings made in the system"
Active Users	"^[0-9]+$" , "Displays number of currently active/registered users"
Pending Returns	"^[0-9]+$" , "Shows number of products yet to be returned"
Total Products	"^[0-9]+$" , "Displays total number of products available in the system"
Available Stock	"^[0-9]+$" , "Shows number of products currently available for rent"
Active Bookings	"^[0-9]+$" , "Displays number of ongoing bookings (not yet completed)
Data Refresh	"Dashboard data should be updated in real-time or on page reload"

6.12 Categories Management

Figure 6.12 shows the category management page where Admin can add, edit, deactivate, and organize product categories and subcategories. A well-organized category structure is essential for the product catalog to be browsable and searchable by customers and staff.

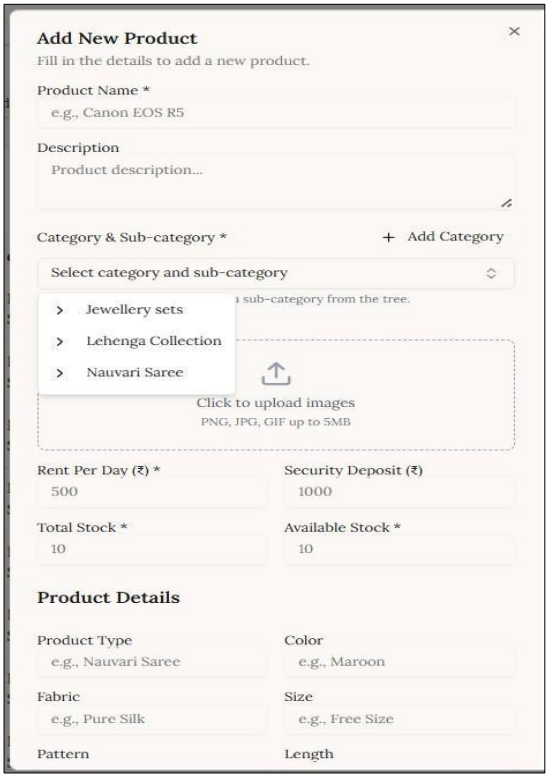


Figure 6.12: Categories Management

6.13 Product Management

Figure 6.13 shows the product management page where Admin can add new rentable products, update existing product details, adjust rental rates, change availability status, and manage product quantities. Each product record stores all the information needed for booking — including its name, associated category, rent price, deposit amount, current status, and available quantity

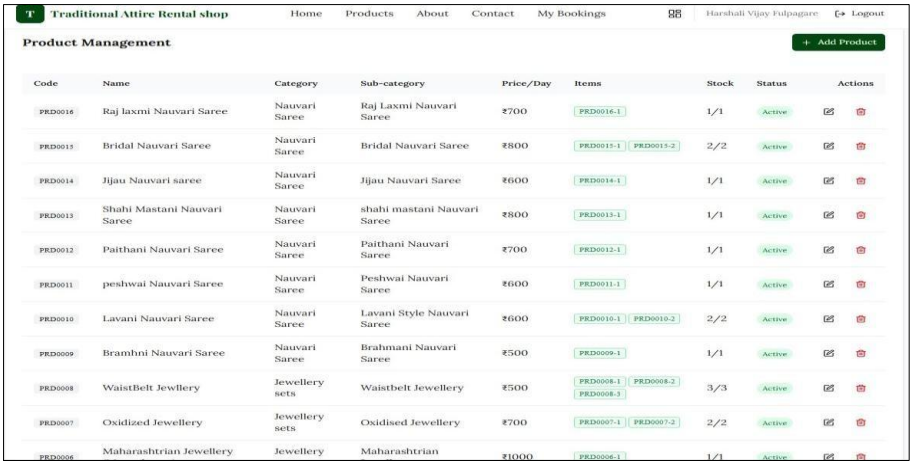


Figure 6.13: Product Management

Main outcomes:

- **Centralized Product Master Control:** All product records are managed from a single page, ensuring consistency across the entire booking workflow.
- **Availability and Rental-Rate Maintenance:** Admin can update rent prices and availability status in real time, keeping the catalog accurate for customers and staff.
- **Data Consistency Across Booking Workflow:** Since all bookings reference the product master, any update to a product record is automatically reflected in related records.

6.14 Customer Management

Figure 6.14 shows the customer records page where authorized users can view, add, update, and manage customer profiles. Each customer record stores identity and contact details including name, phone number, email address, and physical address. These records are used to link customers with their bookings, payments, and rental history.

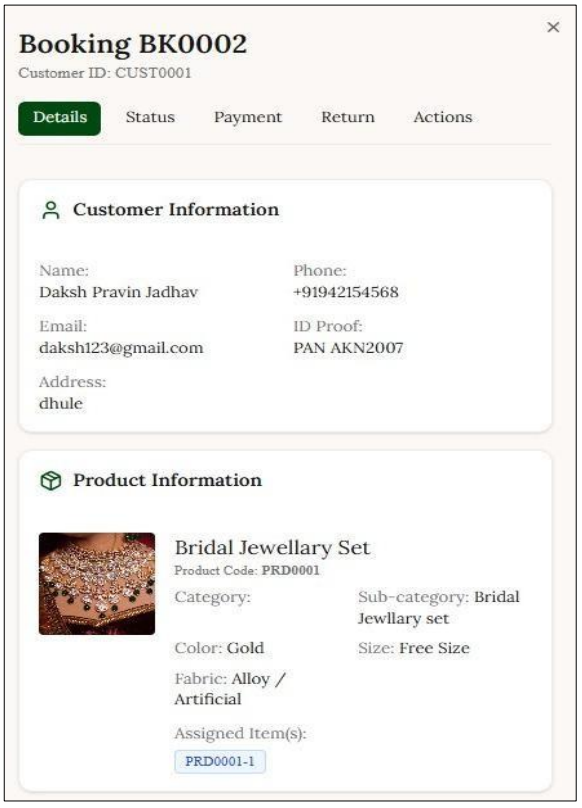


Figure 6.14: Customer Management

Main outcomes:

- **Better Customer Profile Management:** Customer details are stored centrally and can be updated or retrieved quickly during any operational task.
- **Clean Mapping of Customer with Bookings and Payments:** Each booking and payment is linked to a specific customer record, providing a full history
- view per customer.
- **Reduced Duplication and Missing Records:** The structured customer database prevents duplicate entries and ensures that no customer interaction is recorded without a valid customer reference.

6.15 Booking Management (Admin / Staff)

Figure 6.15 shows the operational booking management page used by Admin and Staff. This page lists all bookings in the system with filters for status, date range, and customer. Staff can open any booking to update its status, confirm it, mark it as active when the product is issued, or process it for return when the customer brings back the product.

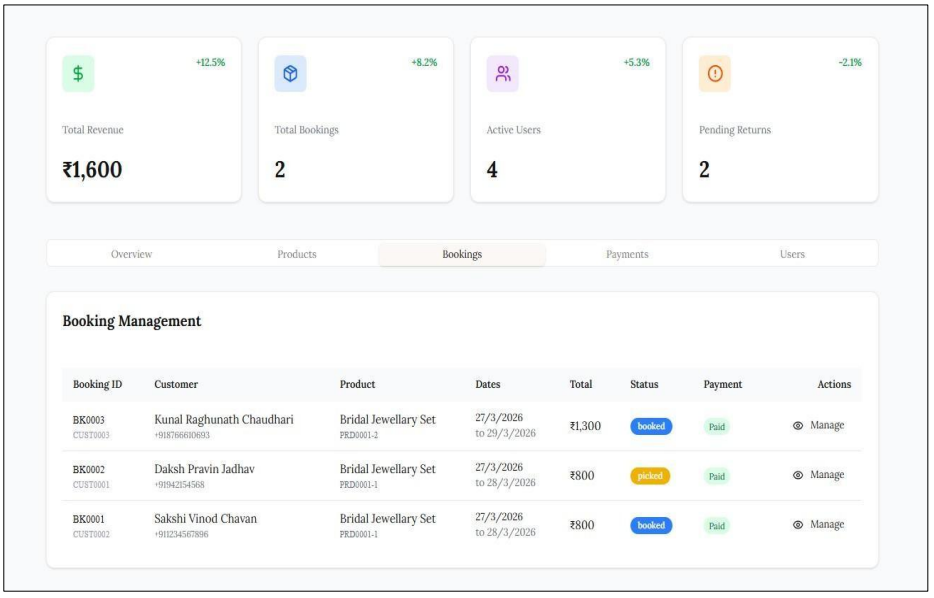


Figure 6.15: Booking Management

Table 6.10: Booking Management Page Validations – Admin Control

Fields	Validation Applied
Booking ID	"Unique ID generated by system; cannot be duplicated or edited"
Customer Name	"Displays customer name linked to bookin should not be empty"
Product Name	"Displays booked product correctly; must exist in product database"
Booking Date	"Shows date of booking; auto-generated and read-only"
Total Amount	"^[0-9]+(\.[0-9]{1,2})?$" , "Must be a positive value; calculated based on bookin duration"
Payment Status	"Shows payment state (Paid / Pending / Failed); must reflect actual transaction"
Action (View/Edit/Delete)	"Admin can view details, update status, or cancel/delete booking with confirmation"

6.16 Inspection Management

Figure 6.16 shows the inspection records page. Inspections are recorded at two key points in the rental lifecycle — before the product is issued to the customer (pre-issue inspection) and after the product is returned (post-return inspection). Each inspection record stores the product condition, inspection date, and any remarks noted by the staff member. This two-point inspection process is the primary mechanism for protecting the rental business from loss due to damage

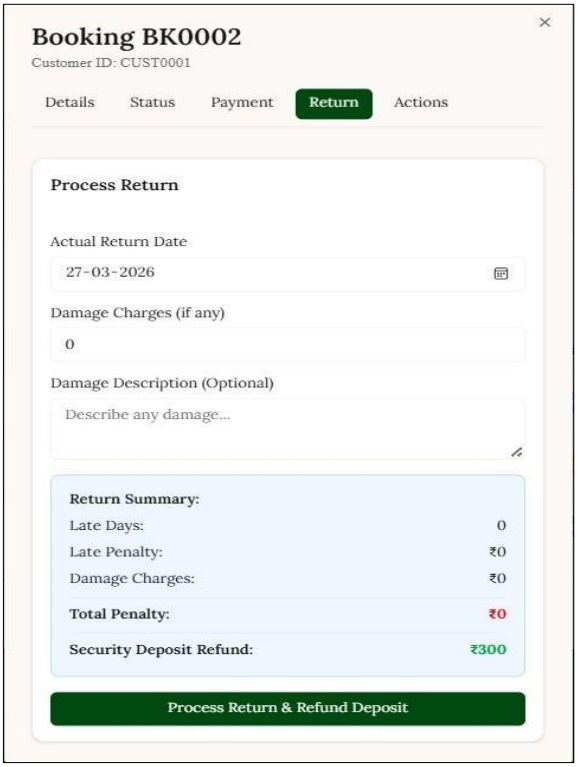


Figure 6.16: Inspection Management

Table 6.11 Inspection Management Validation — Pre-Issue and Post-Return Condition Records

Fields	Validation Applied
Actual Return Date	"Must be current or past date; cannot be before booking start date"
Late Days	"^[0-9]+$" , "Auto-calculated if return date exceeds end date; must be non-negative"
Penalty (Late Fee)	"^[0-9]+(\.[0-9]{1,2})?$" , "Calculated based on late days; must be positive value"
Damage Charges	"^[0-9]+(\.[0-9]{1,2})?$" , "Applicable if product is damaged; must be non-negative"

Total Charges	"^[0-9]+(\.[0-9]{1,2})?$" , "Sum of penalty and damage charges; auto-calculated"

6.17 Return Management

Figure 6.17 shows the return processing page. When a customer returns a product, staff opens the relevant booking on this page and records the return date, the condition of the returned product, the number of late days if the return is past the agreed end date, and any extra charges to be applied. The return record is then linked to the booking and triggers the final payment settlement process.

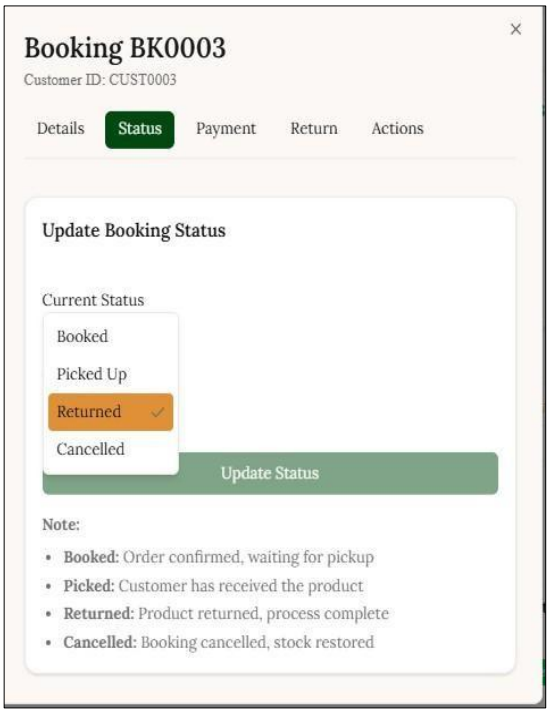


Figure 6.17: Return Management

Table 6.12: Return Management Page Validations – Status Tracking & Updates

Fields	Validation Applied
Booking ID	"Unique booking ID; auto-generated and cannot be edited"
Current Status	"Displays current status: Booked / Picked Up / Returned / Cancelled"

Fields	Validation Applied
Update Status	"Admin can update status only in valid sequence (Booked → Picked Up → Returned / Cancelled)"
Cancellation Rule	"Booking can be cancelled only before 'Picked Up' status"
Action Button (Update)	Enabled only when valid status change is selected

6.18 Payment Management

Figure 6.18 shows the payment management page for recording and tracking all financial transactions related to bookings. Staff can add payment entries here, specifying the amount received, the payment method (cash, UPI, card, etc.), and the transaction reference. The system automatically computes the remaining due amount after each entry.

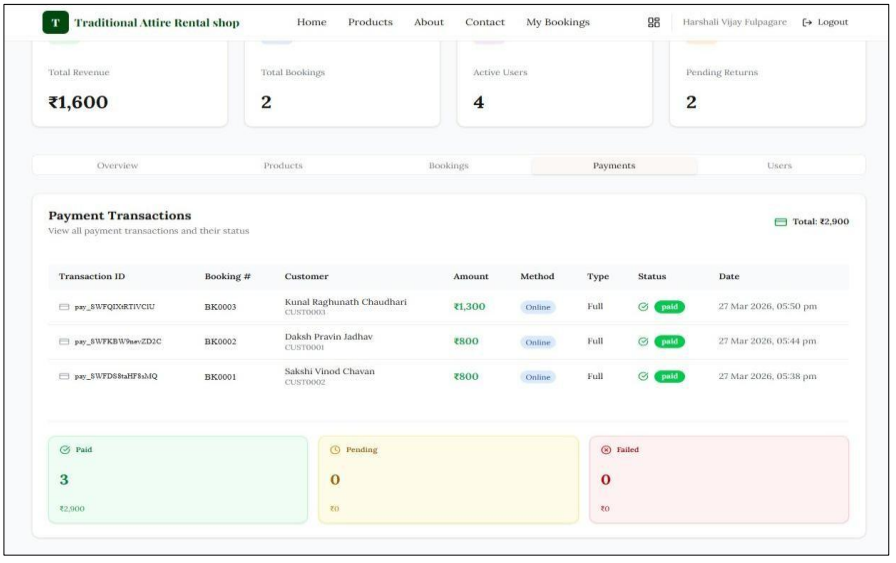


Figure 6.18: Payment Management

Main outcomes:

- **Unified Payment Records:** All payment entries for every booking are stored in one place and linked back to the booking for easy reference.
- **Pending Due Visibility:** The system always displays the current outstanding balance for each booking, making it easy to follow up on unpaid amounts.

- **Better Revenue and Settlement Control:** Admin and Staff can monitor the overall payment health of the business through this page, identifying delays and acting on them promptly.

6.19 User Management

Figure 6.19 shows the user management page where Admin controls system accounts and their assigned roles. Admin can add new users, assign them a role (Admin, Staff, or Customer), update their details, or deactivate accounts that are no longer required. Only Admin has access to this page — neither Staff nor Customer can view or modify user records.

User management is the foundation of the system's security model. By controlling who has what role, Admin ensures that only authorized individuals can perform sensitive operations such as confirming bookings, recording payments, or generating reports

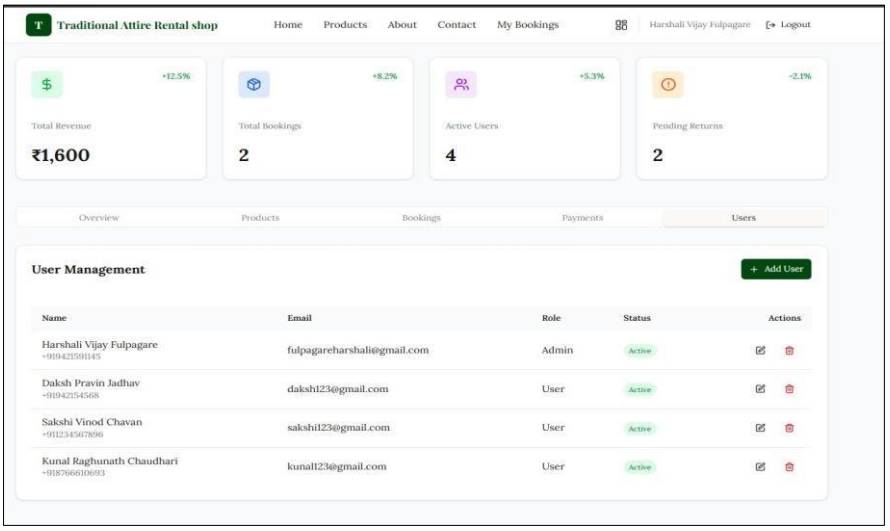


Fig. 6.19: User Management

Main outcomes:

- **Controlled Access Permissions:** Admin can define exactly what each user is allowed to do by assigning the correct role at account creation or any time afterwards.
- **Role-Based Operation Security:** By restricting sensitive operations to specific roles, the system prevents unauthorized actions and protects operational data.

- **Better Accountability in System Usage:** Every system action is tied to a logged-in user, creating a traceable record of who did what and when across all modules.

This chapter demonstrated the effectiveness of the developed system in improving rental operations. The results show that the system reduces manual errors, enhances efficiency, and provides better data management. The application of the system proves its usefulness in real-world rental businesses.

Chapter 7

Testing

This chapter presents the testing and validation process carried out for the Traditional Attire Rental Management System. The primary aim of testing is to ensure that the system functions correctly, efficiently, and meets all specified requirements.

Testing was performed on two major modules of the system: Login Functionality and Product Booking (Core Functionality). The login module acts as the entry point for users and ensures secure authentication, while the product booking module handles the main operations of selecting and booking products.

The objective of this testing phase is to verify that both modules work accurately for valid inputs and appropriately handle invalid inputs. The system was tested for various scenarios including successful login, invalid credentials, empty fields, valid booking, incorrect booking details, and validation checks.

Both positive and negative testing approaches were used to ensure the reliability and robustness of the system. Positive testing verifies correct behavior with valid inputs, whereas negative testing checks how the system responds to incorrect or unexpected inputs.

Testing was carried out using manual methods as well as automation tools such as Selenium IDE and Selenium WebDriver to improve efficiency and accuracy.

7.1 Test Plan

The test plan defines the objectives, scope, testing environment, tools, entry and exit criteria, and risk factors associated with the testing cycle of the Traditional Attire Rental Shop Website.

This test plan has been prepared following the IEEE standard format to ensure a well-structured and standardized documentation

approach. A structured test plan ensures that testing activities are disciplined, consistent, and reproducible, and are aligned with the overall quality goals of the project.

Test Plan: (Traditional Attire Rental Shop Website)

Version:1.0
Test Plan ID: TP_TARSW_01

1. Introduction

This test plan verifies that the Login and Product Booking modules of the Traditional Attire Rental Shop Website work correctly. It ensures that valid users can log in successfully, invalid inputs are handled properly, and product booking operations such as searching, filtering, and selecting products function accurately. It also validates system reliability, usability, and security.

2. Test Items

- Login form UI elements – input fields, submit button, error messages
- Email/username and password validation logic
- Login button action and authentication process
- Product listing and display
- Product search and filtering functionality
- Product selection and booking process

3. Features to be Tested

- Valid and invalid login scenarios
- Empty field validation
- Invalid email format handling
- Forgot password functionality
- Product search functionality
- Category-based filtering

- Product booking process

4. Features Not to be Tested

- Payment module
- Inventory management (admin side)

5. Approach

- Functional and non-functional testing will be performed
- Both positive and negative test cases will be executed
- Manual testing and Selenium automation will be used
- UI validation and usability testing will be included

6. Pass / Fail Criteria

- All high-priority test cases must pass successfully
- Minimum 95%–99% requirement coverage should be achieved
- System should not have any critical or blocking defects
- If blocking defects are found, testing will be stopped and resumed after fixing
- Expected results should match actual results
- All major defects must be fixed and retested before completion

7. Suspension Criteria and Resumption Requirements

- Testing will be stopped if critical bugs occur
- Testing will resume after bug fixing and system stabilization

8. Test Deliverables

- Test plan document
- Test cases document
- Defect report
- Test summary report

- Automation scripts

9. Test Tasks

- Prepare test plan and test cases
- Execute test cases
- Identify and log defects
- Retest fixed defects
- Prepare final report

10. Test Environment

- Operating System: Windows 10/11
- Browser: Google Chrome, Microsoft Edge
- Frontend: React (Vite)
- Backend: Node.js , Express.js
- Database: MongoDB
- Server: Localhost
- Tools: Selenium IDE, Selenium WebDriver

11. Responsibilities

- Test Engineer: Execute test cases and report defects
- Developer: Fix defects
- Project Guide: Review testing process

12. Scheduling

Testing activities are performed during the final phase of project development.

13. Risks

- Incomplete requirements
- Browser compatibility issues

14. Approvals

Project Guide and Team Members

7.2 Test Cases

Test Cases: Login Module

Table 7.1: Test Cases for Login Functionality

Test Case ID	Objective	Input (Steps)	Expected Result	Actual Result	Status
TC_ LGN _001	Verify login with valid admin credentials	Email: fulpagare-harshali@-gmail.com, Password: harshali	User should login successfully and redirect to Admin Dashboard	User logged in and redirected to Admin Dashboard	Pass
TC_ LGN _002	Verify login with valid customer credentials	Email: daksh123-@gmail.com, Password: daksh1	User should login successfully and redirect Home Page	User log in and redirected Home Page	Pass
TC_ LGN _003	Verify login with invalid password	Email: fulpagare-harshali@-gmail.com, Password: harshali	Show "Invalid credentials" and deny login	Error message displayed and login blocked	Pass
TC_ LGN	Verify login with invalid	Email: wrong@-gmail.com,	System should display	Error message displayed	Pass

Test Case ID	Objective	Input (Steps)	Expected Result	Actual Result	Status
_004	email	Password: harshali	error message		
TC_ LGN _005	Verify login with empty fields	Email: blank, Password: blank	System should display "Please fill all fields"	Message displayed	Pass
TC_ LGN _006	Verify login with empty password	Email: fulpagare- harshali@- gmail.com, Password: blank	Show "Password is required"	Message displayed	Pass
TC_ LGN _007	Verify login with empty email	Email: blank, Password: harshali	Show "Email is required"	Message displayed	Pass
TC_ LGN _008	Verify login with invalid email format	Email: fulpagare- harshali, Password: harshali	Show "Invalid email format"	Message displayed	Pass
TC_ LGN _009	Verify forgot password functionality	Click forgot password + enter email	System should send reset link	Reset link not sent	Fail

Test Case ID	Objective	Input (Steps)	Expected Result	Actual Result	Status
TC_ LGN _010	Verify role-based redirection	Login as customer	Redirect to Home Page based on role	Redirect incorrect	Fail

Test Cases: Product Booking Module

Table 7.2: Test Cases for Product Page Functionality

Test Case ID	Objective	Input (Steps)	Expected Result	Actual Result	Status
TC_BK _001	Verify product listing display	Navigate to Product page	All available products should be displayed with details	Products displayed correctly	Pass
TC_BK _002	Verify product search function-ality	Enter valid keyword in search bar	System should display products matching the search keyword	Relevant products displayed	Pass
TC_BK _003	Verify search with no matching results	Enter invalid / random keyword	System display "No productfoun d" message	Message displayed correctly	Pass

Test Case ID	Objective	Input (Steps)	Expected Result	Actual Result	Status
TC_BK_004	Verify category-based filtering	Select category checkbox	Only products of selected category should be displayed	Filter applied correctly	Pass
TC_BK_005	Verify multiple category filtering	Select multiple categories	Products from selected categories should be displayed	Multiple filter working correctly	Pass
TC_BK_006	Verify price range filtering	Adjust price range slider	Products within selected price range should be displayed	Price filter working correctly	Pass
TC_BK_007	Verify sorting (Low to High)	Select "Price Low to High" option	Products should be sorted in ascending order of price	Sorting applied correctly	Pass
TC_BK_008	Verify sorting (High to	Select "Price High to Low"	Products should be sorted in	Sorting applied	Pass

Test Case ID	Objective	Input (Steps)	Expected Result	Actual Result	Status
	Low)	option	descending order of price	correctly	
TC_BK_009	Verify sorting by name	Select "Name (A-Z)" option	Products should be sorted alphabetically	Sorting applied correctly	Pass
TC_BK_010	Verify product selection	Click on any product	Selected product details page should open	Product details displayed	Pass
TC_BK_011	Verify filter reset functionality	Click on "Clear Filters" button	All filters should be removed and all products displayed	Filters cleared successfully	Pass
TC_BK_012	Verify combined filters	Apply search + category + price filter	System display products matching all conditions	Combined filtering works correctly	Pass
TC_BK_013	Verify search with special	Enter "@@@### " in search	System should show "No	System shows irrelevant	Fail

Test Case ID	Objective	Input (Steps)	Expected Result	Actual Result	Status
	characters	bar	products found" or handle input gracefully	products or crashes	

7.3 Defect Report

The defect report is a crucial document used to record and track defects identified during the testing process of the Traditional Attire Rental Shop Website. It provides detailed information about each defect, including its description, severity, priority, steps to reproduce, and current status.

This report helps the development and testing teams to understand, prioritize, and resolve issues efficiently. It also ensures effective communication among team members and maintains transparency throughout the defect management process.

Defect Report (Login – Forgot Password)

Table 7.3: Defect Report for Login – Forgot Password

Defect ID	DR_LOGIN_001
Project Name	Traditional Attire Rental Shop Website
Product	Web Application
Release Version	v1.0
Module	Login Module – Forgot Password
Detected Build	v1.1
Summary	Reset password link is not sent to registered email
Description	While testing the forgot password functionality, it was observed that when a valid registered email is entered, the system does not send any password reset link. The

	system also does not display any success or error message, which prevents the user from resetting the password.
Steps to Reproduce	1. Open login page 2. Click on "Forgot Password" 3. Enter valid registered email 4. Click on submit/reset button
Expected Result	System should send a password reset link to the registered email
Actual Result	No reset link is sent and no message is displayed
Severity	High
Priority	High
Reported By	Test Engineer
Status	Open

Defect Report (Login – Role-Based Redirection)

Table 7.4: Defect Report for Login – Role-Based Redirection

Defect ID	DR_LOGIN_002
Project Name	Traditional Attire Rental Shop Website
Product	Web Application
Release Version	v1.0
Module	Login Module – Role-Based Redirection
Detected Build	v1.1
Summary	Incorrect redirection after login based on user role
Description	While testing role-based redirection, it was observed that after successful login using customer credentials, the system does not redirect the user to the correct Home Page. The role validation logic is not properly implemented, resulting in incorrect navigation.

Steps to Reproduce	1. Open login page 2. Enter valid customer credentials 3. Click on login button
Expected Result	User should be redirected to the Home Page based on role
Actual Result	User is redirected incorrectly or role is not validated properly
Severity	Medium
Priority	High
Reported By	Test Engineer
Status	Open

Defect Report (Product Booking – Search Issue)

Table 7.5: Defect Report for Product Booking – Search Issue

Defect ID	DR_BK_001
Project Name	Traditional Attire Rental Shop Website
Product	Web Application
Release Version	v1.0
Module	Product Booking – Search Functionality
Detected Build	v1.1
Summary	Incorrect behavior when special characters are used in search
Description	While testing the search functionality, it was observed that entering special characters such as "@@@###" does not return appropriate results. The system either shows irrelevant products or behaves unexpectedly, instead of handling the input properly.
Steps to Reproduce	11. Open product page 2. Enter "@@@###" in search bar 3. Press search
Expected Result	System should display "No products found" or handle input properly

Actual Result	System shows irrelevant products or crashes
Severity	Medium
Priority	High
Reported By	Test Engineer
Status	Open

7.4 Automation Testing (Selenium)

Automation testing is performed to validate the functionality of the Traditional Attire Rental Shop Website using automated tools. It helps in reducing manual effort, improving accuracy, and ensuring consistency in repetitive test execution.

In this project, automation testing is carried out using Selenium IDE and Selenium WebDriver. These tools are used to automate the testing of Login and Product Booking modules.

Selenium IDE is a record-and-playback tool that allows testers to create automated test cases without programming knowledge. It is mainly used to record user actions such as entering login credentials, clicking buttons, and navigating between pages. The recorded test scripts can be executed multiple times to verify system behavior.

Selenium WebDriver is an advanced automation testing tool that allows interaction with web browsers through programming. It is used to write scripts that simulate user actions such as typing input, clicking elements, and validating outputs. WebDriver supports multiple browsers and provides better control for handling complex test scenarios.

Automation was performed by recording login scenarios such as valid login, invalid login, and empty field validation using Selenium IDE. Similarly, product booking functionalities such as search, filtering, and product selection were also automated.

The automated test scripts were executed multiple times to ensure consistent results. The results obtained from automation testing matched the expected outcomes for most test cases, except for certain failed scenarios such as forgot password functionality and incorrect search behavior.

7.4.1 Selenium IDE Execution

In this project, Selenium IDE was used to automate the Login and Product Booking functionalities. The recorded scripts were executed multiple times to verify consistent system behavior. The execution results showed that most of the test cases passed successfully, while some scenarios such as forgot password functionality failed.

Login Functionalities Selenium :-

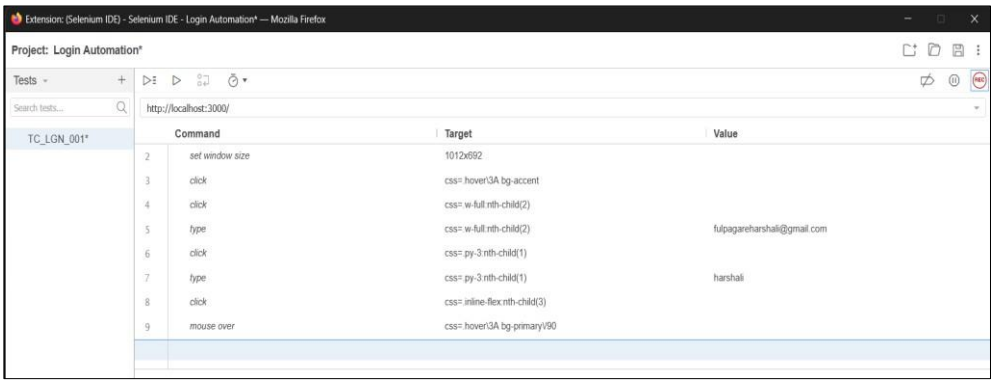


Figure 7.1: Valid Login (Valid Email & Valid Password)

This screenshot shows the execution of the login test case using valid email and valid password as per the decision table combination. The automation script enters correct credentials and performs the login action. The system successfully authenticates the user and redirects to the dashboard. This confirms that the system behaves correctly for valid inputs and allows authorized access.

Figure 7.2: Invalid Password (Valid Email & Invalid Password)

This screenshot illustrates the execution of the login test case with a valid email and an invalid password. The automation script attempts to log in using incorrect password credentials. The system detects the mismatch and displays an appropriate error message such as "Invalid password" while preventing access. This verifies proper password validation.

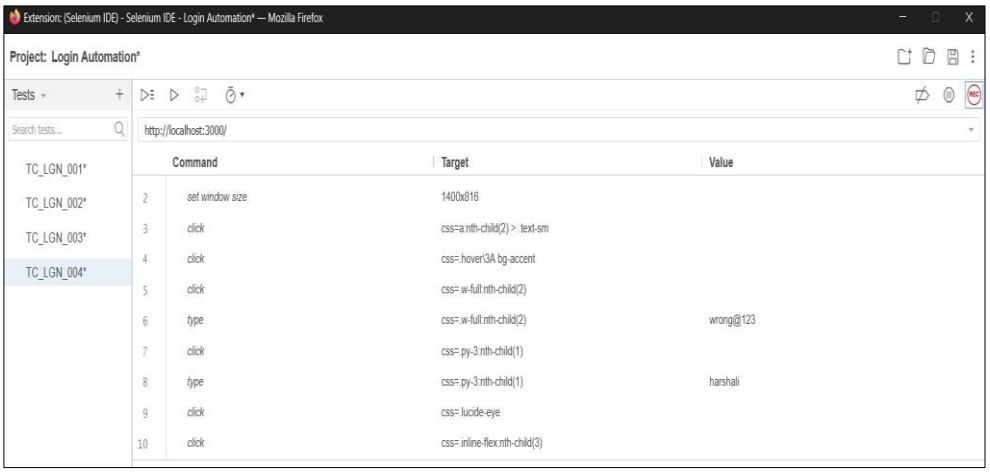


Figure 7.3: Invalid Email (Invalid Email & Valid Password)

This screenshot represents the execution of the login test case with an invalid email and a valid password. The system identifies that the email is not registered or incorrect and displays an error message. The login attempt is denied, ensuring that only valid users can access the system.

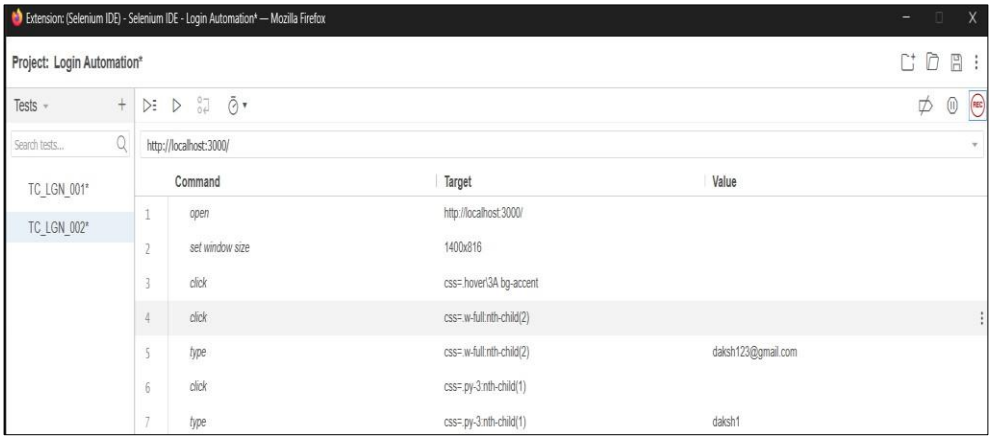


Figure 7.4: Invalid Credentials (Invalid Email & Invalid Password)

This screenshot shows the execution of the login test case where both email and password are invalid. The system displays a general error message such as "Invalid credentials" and blocks the login attempt. This confirms that

the system properly handles incorrect input combinations and maintains security.

Product Page Functionalities Selenium:-

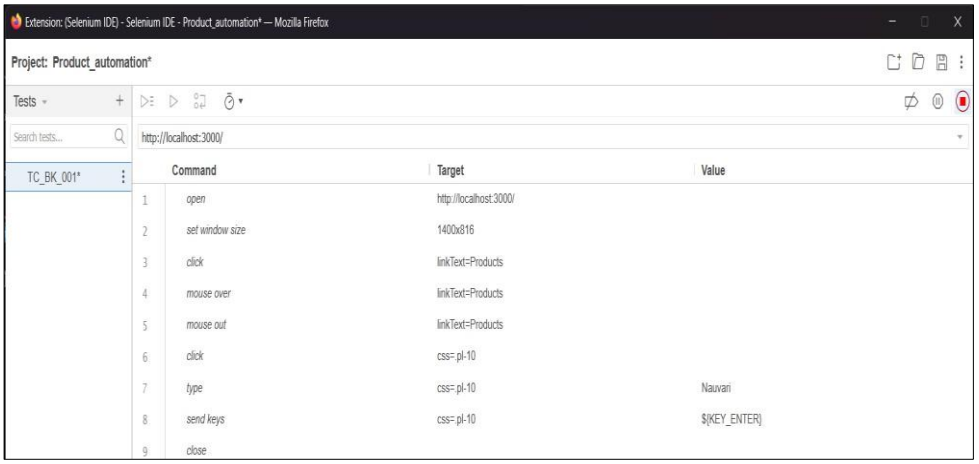


Figure 7.5: Valid Search without Filter

This screenshot shows the execution of product search functionality using a valid keyword without applying any filters. The automation script enters a valid product name in the search bar and displays all matching products. This confirms that the search functionality works correctly and returns relevant results based on user input.

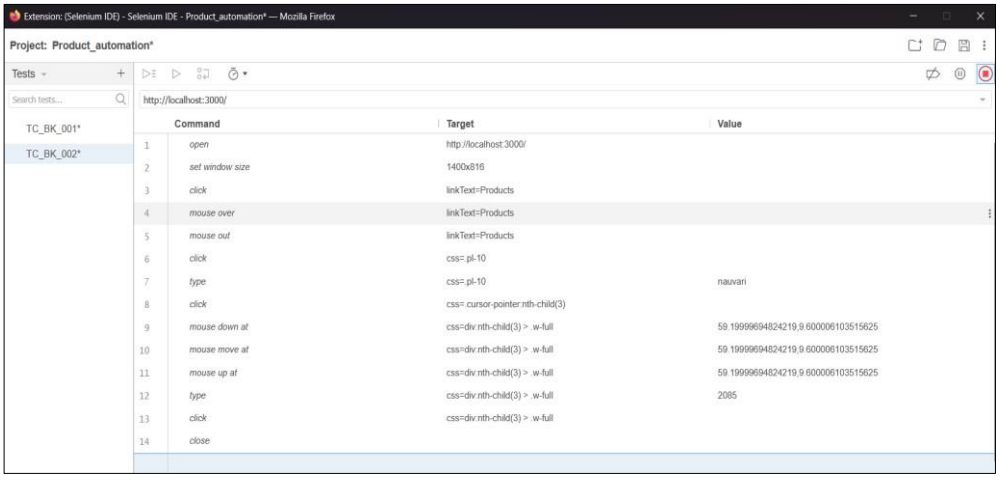


Figure 7.6: Valid Search with Filter Applied

This screenshot illustrates the execution of product search combined with filtering. The system applies both search input and category/price filter conditions. Only products matching both criteria are displayed. This verifies

that the system correctly handles multiple conditions and displays accurate filtered results.

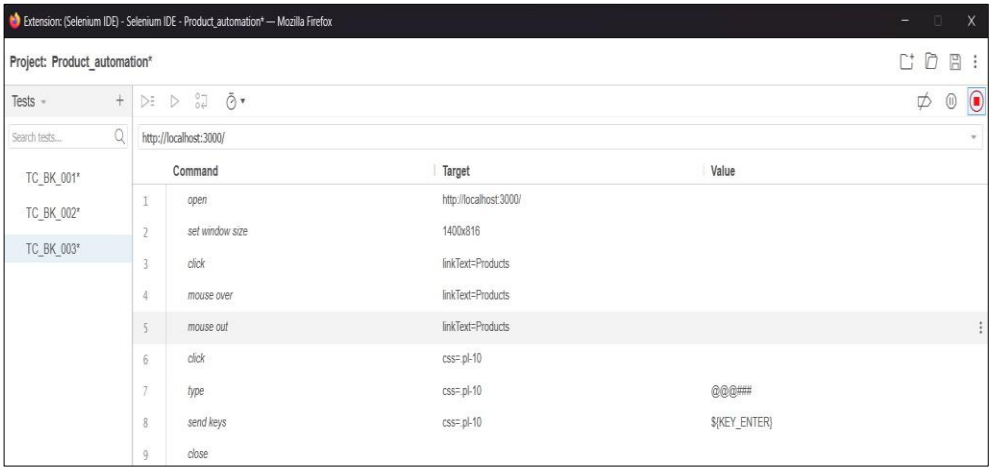


Figure 7.7: Invalid Search without Filter

This screenshot represents the execution of search functionality using an invalid or random keyword without applying any filters. The system does not find any matching products and displays a message such as "No products found". This confirms proper handling of invalid search inputs.

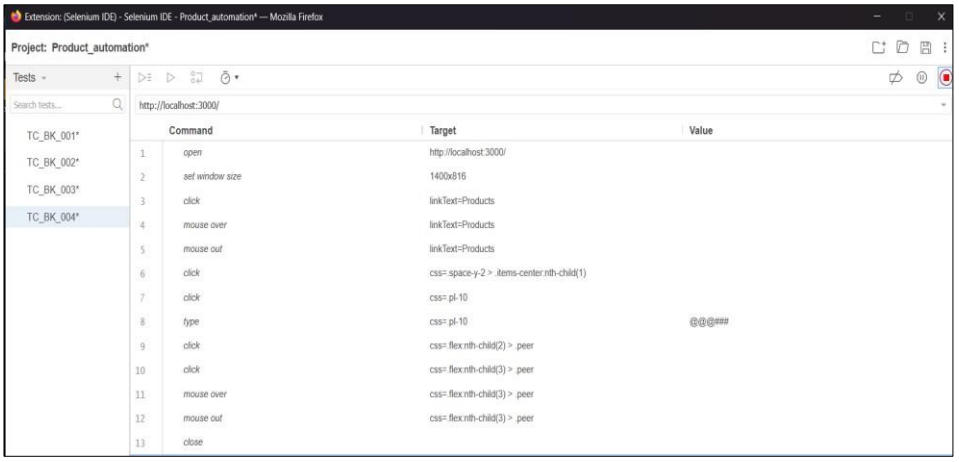


Figure 7.8: Invalid Search with Filter Applied

This screenshot shows the execution of search with invalid input along with applied filters. The system evaluates both conditions and displays no results along with an appropriate message. This verifies that the system correctly handles edge cases and combined invalid conditions.

7.4.2 Selenium WebDriver Execution

Selenium WebDriver is used to automate testing by interacting directly with the web browser using scripts. It simulates user actions such as entering inputs, clicking buttons, and navigating between pages.In this project, WebDriver was used to automate both Login and Product Booking functionalities. It provides better control and flexibility compared to Selenium IDE.

For the Login module, WebDriver automates entering email and password, clicking the login button, and verifying successful login or error messages. It also checks role-based redirection.

For the Product Booking module, WebDriver automates product search, filtering, and validation of displayed results. It ensures correct behavior for both valid and invalid inputs.

WebDriver Script (Login):-

```
import org.openqa.selenium.By;

import org.openqa.selenium.WebDriver;

import org.openqa.selenium.chrome.ChromeDriver;

public class LoginTestCases {

    public static void main(String[] args) throws
        InterruptedException {

        WebDriver driver = new ChromeDriver();

        driver.manage().window().maximize();

        // Open Login Page

        driver.get("http://localhost:3000/login");

        Thread.sleep(2000);

        //========= TC_LGN_001 ============
```

```
System.out.println("TC_LGN_001 - Valid Customer
Login");
driver.findElement(By.name("email")).clear();
driver.findElement(By.name("email")).sendKeys("fulpagareh
arshali@gmail.com");
driver.findElement(By.name("password")).clear();
driver.findElement(By.name("password")).sendKeys("harshali");
driver.findElement(By.name("login")).click();
Thread.sleep(3000);
// ============ TC_LGN_002 ============
System.out.println("TC_LGN_002 - Invalid Password");
driver.get("http://localhost:3000/login");
Thread.sleep(2000);
driver.findElement(By.name("email")).clear();
driver.findElement(By.name("email")).sendKeys("fulpagareh
arshali@gmail.com");
driver.findElement(By.name("password")).clear();
driver.findElement(By.name("password")).sendKeys("wron
g123");
driver.findElement(By.name("login")).click();
Thread.sleep(2000);
driver.quit();
}
}
```

Execution for login:-

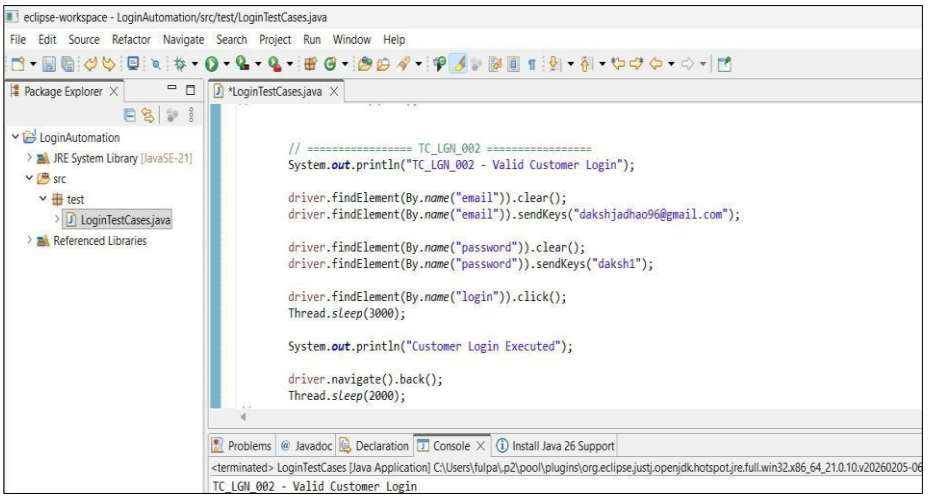


Figure 7.9 Valid Customer Login

This screenshot shows the execution of the Selenium WebDriver script for valid customer login. The script enters correct customer email and password and performs the login action. The system successfully authenticates the user and redirects to the Home Page. This confirms correct login functionality and role-based navigation.

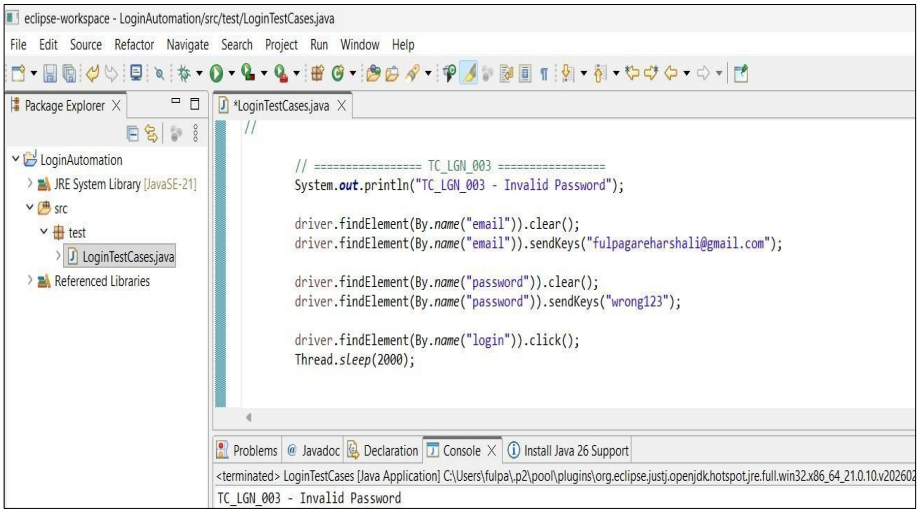


Figure 7.10 Invalid Password

This screenshot shows the execution of the Selenium WebDriver script for valid customer login. The script enters valid customer credentials and performs the login action. The system successfully logs in the user and redirects to the Home Page based on the user role. This verifies correct role-based authentication and navigation.

WebDriver Script (Product Page):-

```
import org.openqa.selenium.By;

import org.openqa.selenium.WebDriver;

import org.openqa.selenium.chrome.ChromeDriver;

public class ProductTest {

    public static void main(String[] args) throws InterruptedException {

        WebDriver driver = new ChromeDriver();

        driver.manage().window().maximize();

        driver.get("http://localhost:3000/products");

        Thread.sleep(2000);

        // ================= TC_BK_001 =================

        System.out.println("TC_BK_001 - Valid Search");

        driver.findElement(By.xpath("//input[@placeholder='Search
          products...']")) .sendKeys("Nauvari");

        Thread.sleep(3000);

        // ================= TC_BK_002 =================

        System.out.println("TC_BK_002 - Invalid Search");

        driver.findElement(By.xpath("//input[@placeholder='Search
          products...']")).clear();

        driver.findElement(By.xpath("//input[@placeholder='Search
          products...']")) .sendKeys("@@@###");

        Thread.sleep(3000);

        driver.quit();

    }

}
```

Execution for Product Page:-

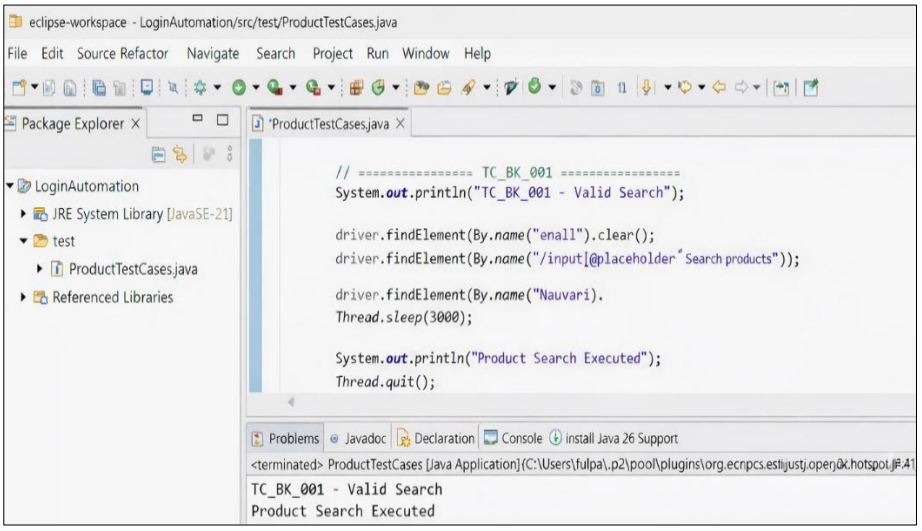


Figure 7.11: Valid Product Search

This screenshot shows the execution of the product search functionality using Selenium WebDriver with a valid keyword. The system displays relevant products based on the search input, confirming correct working of the search feature

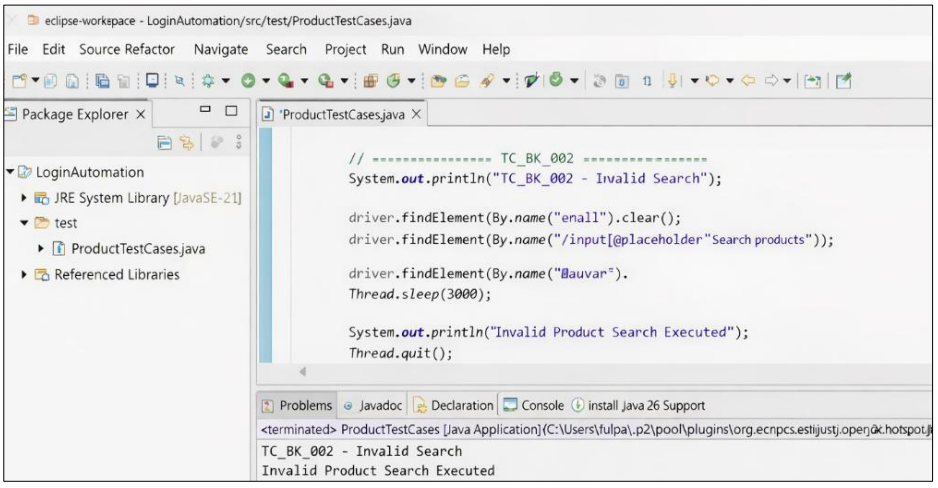


Figure 7.12: Invalid Product Search

This screenshot shows the execution of the product search with an invalid input. The system does not display any matching products and shows a message such as "No products found". This confirms proper handling of invalid search inputs.

7.5 Test Summary

The test summary provides an overview of the testing activities performed on the Traditional Attire Rental Shop Website. The testing process was carried out on two main modules: Login Functionality and Product Booking.

A total of multiple test cases were designed and executed to validate both modules. The test cases covered various scenarios including valid inputs, invalid inputs, boundary conditions, and user interface validations. Both positive and negative testing approaches were used to ensure system reliability and robustness.

The majority of test cases were executed successfully, and the system performed as expected under normal conditions. The login module successfully authenticated valid users and handled invalid inputs with appropriate error messages. Similarly, the product booking module correctly displayed products, supported search, filtering, and sorting functionalities.

However, some test cases failed during execution. Issues were identified in functionalities such as forgot password and role-based redirection in the login module, and search behavior with special characters in the product booking module. These defects were documented and reported for further resolution.

Automation testing using Selenium IDE and Selenium WebDriver was also performed to validate repetitive test scenarios. Automation helped in improving testing efficiency and ensured consistent execution of test cases.

Overall, the system meets most of the functional requirements and performs reliably. The identified defects are manageable and can be resolved to further enhance system performance and user experience. The system is considered stable and ready for deployment after fixing the reported issue.

Chapter 8

Conclusion and Future Scope

8.1 Conclusion

The Traditional Attire Rental Management System has been successfully designed and implemented to transform manual rental operations into a structured and efficient digital platform. The system integrates all major rental activities including product management, booking, inspection, return handling, payment tracking, and reporting into a centralized workflow.

The application effectively supports all key users in the rental business. The Admin is provided with complete control over users, categories, products, and system reports. Staff members are able to perform operational tasks such as checking availability, managing bookings, handling returns, and updating payments efficiently. Customers are given a simple and user-friendly interface to browse attire, place booking requests, and track their booking status.

The system enhances overall performance by ensuring real- time product availability, reducing booking conflicts, and maintaining accurate records of all transactions. The structured booking lifecycle and proper inspection handling help in minimizing disputes and improving accountability.

The testing results confirm that the system performs reliably under various scenarios, with most functionalities working as expected and only minor improvements required.

In conclusion, the developed system provides a reliable, efficient, and scalable solution for traditional attire rental businesses. It reduces manual workload, improves accuracy, enhances customer experience, and supports future growth through digital transformation.

8.2 Future Scope

The current system establishes a strong foundation for rental operations. In future versions, the following enhancements can further improve performance and user experience:

- **Instant Notifications:** Send automatic alerts through SMS, WhatsApp, and email for booking confirmation, return reminders, overdue status, and payment updates.
- **Digital Verification and e-Signature:** Add customer document verification and digital signature support during booking and issue/return handover.
- **QR or Barcode Integration:** Use barcode or QR scanning to speed up attire issue, return check-in, and inventory identification.
- **Mobile Application:** Provide Android or cross-platform mobile access for staff and customers to manage operations from anywhere.
- **Online Payment Gateway Integration:** Support integrated UPI, card, and wallet payments with automated receipt generation and reconciliation.
- **Smart Forecasting:** Use historical booking patterns to predict demand, optimize inventory utilization, and recommend procurement decisions.
- **Advanced Analytics Dashboard:** Add trend-based analytics such as peak rental periods, high-performing attire, delayed return ratios, and customer behavior insights.
- **Multi-Branch Support:** Expand the architecture to support branch-level stock, branch transfers, and centralized monitoring.
- **Audit and Compliance Logs:** Strengthen governance with detailed activity logs, approval workflows, and exportable reports.
- **Recommendation Engine:** Suggest alternative attire or rental durations when requested items are unavailable.

Chapter 9

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